Why do scaling laws in recursive composition keep taking the same few mathematical forms?

Theory · Framework paper

Recursive Dynamics · The ARC Theory · Law I, the ARC Principle, a law under test

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Abstract

If the way a process combines its steps fixes the family of forms its growth can take, then the family a self-improving system's capability will follow as its recursion deepens can be read from how it composes, before it scales and while correction can still be built into it. This paper asks why scaling laws in recursive composition keep taking the same few mathematical forms. Where a process has a well defined composition operator on its input and output and meets the regularity conditions that Cauchy's functional equations require, the admissible forms fall into four families: linear, power law, exponential and logarithmic. This is a conditional classification, not a survey of nature, and saturation lies outside it. Within the power-law family, the exponent $\alpha = d/(d+1)$ for a d-dimensional hierarchical network is earlier work's; this paper gives an account of why those derivations, from different assumptions, converge, and a proof that its three conditions suffice does not yet exist. The relation is read as a network-partition identity with a stated domain: published exponents for filamentous fungi ($d = 1$) do not reject it (a mean of three factors on different axes; no equivalence shown), those for vertebrates and invertebrates with three-dimensional transport ($d = 3$) agree with it on the group means compiled here, though the mammalian exponent is debated, and no organism with a two-dimensional hierarchical network has been identified. Under $d = 2/(1+3w)$ the Friedmann solutions take the same form, an identity that carries no evidence. For finite d the exponent stays below one, a bound on systems of that class only; how capability grows with recursive depth is a different pair of variables, which this geometry does not fix and which has to be measured. Five predictions are stated; one, on neural scaling exponents, is withdrawn because published measurements refute it.

Version 3.13 sets the paper in this form and names its domain, recursive composition, in its title; it withdraws Prediction 2, brings the body to the scope correction of 22 August 2026 and to the corrections owed since, and states Laws IV and V as registered on 15 October 2026; Appendix A lists every version and correction, with its date.

A framework paper of the ARC Theory, within Recursive Dynamics: it traces the scaling forms of Law I, the ARC Principle, a law under test, to their mathematical origin and compares them with published values in biology, physics and cosmology; the $d/(d+1)$ formula is earlier work's, credited in Section 2, and the account of why those derivations converge is this paper's. It also states a prediction about the Eden Protocol and reports a pilot of it, and it decides none of P15, P12 and P21, the propositions of Laws IV and V (registered as laws on 15 October 2026 in version 1.103 of the theory registration; Section 16 gives their status). Working paper, version 3.13, 31 October 2026; first published 22 February 2026; Appendix A lists every version and correction. Recorded on OSF as *Framework Paper · The ARC Theory · Law I, the ARC Principle · On the Origin of Scaling Laws in Recursive Composition: Functional Equations and the Scaling Forms Behind the Law*, DOI 10.17605/OSF.IO/XZY9U. About the author, founder of Recursive Dynamics, without an institution or a doctorate: michaeldariuseastwood.com/questions/who-is-michael-darius-eastwood; his book, *Infinite Architects* (first edition, 2 January 2026): osf.io/dqm3e. Licence: CC BY 4.0.

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1 Introduction

If the way a process combines its steps fixes the family of forms its growth can take, then how a self-improving system composes tells us, before it scales, which family its growth with recursive depth can follow; its exponent still has to be measured (Section 11). That matters for artificial intelligence. Whether correction can keep pace with a system that improves itself depends on how fast its capability compounds, and correction is best built into such a system before it scales, while people still hold oversight of it. This paper traces the forms behind Law I, the ARC Principle, a law under test, to their mathematical origin, in the systems where scaling laws were first measured: organisms, materials and the expanding universe.

A mouse's heart beats about 600 times per minute. An elephant's beats about 28 (Merck Veterinary Manual 2004). A blue whale's, on a foraging dive, beats 4 to 8 (Goldbogen et al. 2019).

If you plot heart rate against body mass across mammals, the points fall close to a straight line on logarithmic axes, with a slope close to $-\frac{1}{4}$, from the mouse to the whale (Stahl 1967; Noujaim et al. 2004).

Why?

For filamentous fungi, published colony-level measurements put the exponent of energy use against colony size near one half, and this paper predicts exactly $\frac{1}{2}$ for a single thread (Section 14). Not $\frac{3}{4}$, like a mammal. Not $\frac{2}{3}$, like the matter-dominated universe. A different fraction entirely.

Why $\frac{1}{2}$ for fungi, but $\frac{3}{4}$ for mammals? And why do these same fractions appear in cosmology?

Now consider something entirely different. The universe is expanding. During the matter-dominated era, the scale factor grew as time raised to the power $\frac{2}{3}$. During the radiation era, it grew as time to the power $\frac{1}{2}$.

Why $\frac{2}{3}$? Why $\frac{1}{2}$? And why are these the same fractions that appear in biology?

Now consider one more thing. When a rock shatters, the distribution of fragment sizes follows a power law; Section 6.1 gives the status of its exponents.

This paper argues that these questions share one mathematical answer, under conditions it states, and grades where that answer is evidence and where it is only an identity.

The answer has two parts, and the paper grades each. The first is mathematical: where a process has a composition operator and meets regularity conditions, the way its steps combine fixes the family its growth can take, among four (Section 4). The second is the exponent: within the power-law family, the formula $\alpha = d/(d+1)$ for a d-dimensional hierarchical network, which earlier work derived (Section 2), gives $\frac{1}{2}$, $\frac{2}{3}$ and $\frac{3}{4}$, and this paper offers an account of why derivations from different assumptions arrive at it. Where the fractions reappear in cosmology the agreement is an identity under a change of variables, and it carries no evidence (Section 9).

Our contributions are:

- **An account of convergence.** Cauchy's multiplicative functional equation, with space filling and a conservation or optimisation constraint on resource flow, gives the power-law family and the exponent $d/(d+1)$; this is offered as the reason why the earlier derivations, from different assumptions, converge on a formula that is theirs (Section 2). A proof that the three conditions suffice does not yet exist.
- **The classification applied to scaling forms.** The four Cauchy families, linear, power law, exponential and logarithmic, set out as the homomorphisms between the additive and multiplicative structures under regularity, with saturation placed outside them and the variable pair named at each site (Sections 3 to 5).
- **The relation as a network-partition identity.** Its domain and its failures stated, and its comparison with published exponents in biology and physics graded row by row (Sections 6 and 12).
- **The geometric bound, scoped.** The exponent stays below one for every finite d in the class of systems whose growth is carried through a space-filling hierarchical network, with the counterexamples

outside that class conceded (Section 10), and the bound distinguished from the exponent of capability against recursive depth (Section 11).

- **Predictions and their status.** Five predictions, one of them withdrawn because published measurements refute it (Section 14).

Section 2 sets the work against earlier work and states what this paper adds. Sections 3 to 5 set out the patterns, Cauchy's classification and the principle; Section 6 the formula and its domain; Section 7 the heart-rate chain; Section 8 the fission chain reaction, which the grid classifies and does not explain; Section 9 the cosmological identity; Section 10 the bound; and Section 11 the distinction between physical and recursive scaling. Section 12 grades the evidence, Section 13 gathers the equations, Section 14 states the predictions and Section 15 the composition operator's wider reading. Section 16 says what the paper does and does not show, Section 17 states its limitations and Section 18 concludes. Appendix A gives the version history and every correction, Appendix B the priority record, Appendix C the notation and a glossary, Appendix D the paper's place in the ARC Theory and the programme, and Appendix E the paper's novelty statements beside the closest earlier work it names.

2 Related work and what this paper adds

The mathematics of this paper is two hundred years old. The biological data is ninety years old. The cosmological solution is a century old. It is reasonable to ask: what, precisely, is the contribution?

2.1 Earlier work

The $d/(d+1)$ formula itself is not new. It is derived, in its general form or as a special case, in earlier work, and the first entry below gives each work what it states: West, Brown and Enquist (1997) from fractal branching networks, Banavar, Maritan and Rinaldo (1999) and Banavar and co-authors (2002) from geometric constraints on transportation networks, He & Zhang (2004) from fractal cell geometry, Demetrius (2003, 2006) and Demetrius & Tuszynski (2010) from quantum oscillator coupling in the Debye model, Bettencourt (2013) from urban scaling theory, as a special case, Zhao (2022) from producer-consumer dynamics, and Maino et al. (2014) from Dynamic Energy Budget theory. Each derivation starts from different assumptions and arrives at the same functional form. This convergence is itself remarkable and requires explanation.

2.2 What this paper adds

The contribution is the **unification**. One formula, $\alpha = d/(d+1)$, connects fungi, mammals, surface physics and, as an identity, the expansion of the universe, with earthquake energy as a variable-pair case study and rock fracture awaiting sourced values (Section 6.1). One equation with no fitted parameter: d is set by the network's dimension. The claim is not that this formula is original, but that Cauchy's functional equations, combined with the space-filling and conservation conditions, explain *why* the earlier derivations, from different starting assumptions, arrive at the same form. That invariance under rescaling forces a power law, through the multiplicative functional equation, is a standard result (Newman 2005, section III.E); what this paper adds is the account of why the derivations of $d/(d+1)$ meet the space-filling and conservation conditions that select the exponent within that family. The derivation of West et al. (1997) has itself been questioned (Dodds et al. 2001; Kozłowski and Konarzewski 2004), and its three-quarter exponent shown to hold only in the limit of infinite network size (Savage et al. 2008); Section 2.3 states what follows if the convergence reflects shared assumptions.

A formal proof that those conditions are sufficient does not yet exist (see the strongest objection, below). Set against the earlier work, the paper's claims are that it has:

1. **Given an account of why derivations from different assumptions converge on one formula.** That account is what this paper adds; the formula, and the values $\frac{1}{2}$, $\frac{2}{3}$ and $\frac{3}{4}$ that it gives, are the earlier work's. West, Brown and Enquist (1997) derive the three-quarter power law for metabolic rates from a model of transport through space-filling fractal networks of branching tubes, a landmark result, and note that, because the network must fill the entire D-dimensional volume, the result generalises to $D/(D+1)$; West, Brown and Enquist (1999) obtain $D/(D+1)$ in D dimensions again, from a more general model based on the geometry rather than the hydrodynamics of hierarchical networks. Banavar, Maritan and Rinaldo (1999) derive a general relationship between size and flow rates in arbitrary networks with local connectivity, and find that in the most efficient class of networks in D dimensions metabolic rate scales as mass to the power $D/(D+1)$, which gives quarter-power scaling for $D = 3$; Banavar et al. (2010) show, in two models of resource distribution networks, that the exponent of metabolic rate on body mass is $\frac{2}{3}$ if the velocity of flow remains constant and can attain a maximum of $\frac{3}{4}$ if the velocity scales with its maximum exponent, 1/12. Demetrius & Tuszynski (2010) derive the exponent $d/(d+1)$ for the scaling of metabolic rate with body size in the regime they call quantum, where d is the dimension of the physical space in which their network of coupled energy-transducing oscillators is embedded, and state its values $\frac{1}{2}$, $\frac{2}{3}$ and $\frac{3}{4}$ for $d = 1$, 2 and 3; in the regime they call classical the scaling is isometric. Bettencourt (2013) sets out a theoretical framework that predicts the average social, spatial and infrastructural properties of cities as a set of scaling relations. In it the land area of a city scales as its population to the power $D/(D+H)$ in D dimensions, where H is the fractal dimension of travel paths; the exponent is $\frac{2}{3}$ for $D = 2$ and $H = 1$, and it has the form $D/(D+1)$ only when $H = 1$. Zhao (2022) derives the $n/(n+1)$ power scaling law for an n-dimensional system from producer-consumer dynamics. He and Zhang (2004) derive a 4/5 law for the human brain, the value $d/(d+1)$ takes at $d = 4$, from the fractal dimension of a brain cell's contour. Maino et al. (2014) derive from dynamic energy budget theory an equation for metabolic rate against mass, the same equation as the finite-size form of West, Brown and Enquist's model, which converges on the three-quarter power only in the limit of large mass; they stress that under neither framework is metabolic rate a power function of mass. The surface area hypothesis predicts $\frac{2}{3}$ but not $\frac{3}{4}$ or $\frac{1}{2}$. The formula $\alpha = d/(d+1)$ predicts all three from a single equation with one input, the dimension of the hierarchical transport network. The $d = 1$ value is not rejected by published fungal data (a mean of three factors on different axes; no equivalence shown) and the $d = 3$ value agrees with published group means; the $d = 2$ value appears in cosmology as an identity, which carries no evidence (Section 9), and in physics in the rows whose grades Section 6.1 gives; in biology it is untested (Section 12).
2. **Connected biological scaling to physics scaling.** The same formula gives Kleiber's $\frac{3}{4}$ law (biology), KPZ roughness at $\frac{1}{2}$ (surface physics), the specific-heat analogue of two-dimensional percolation at $\frac{2}{3}$ (statistical mechanics) and, once their values are sourced, fragmentation in 2D and 3D (materials science). Earlier work already takes the form beyond organisms: Banavar, Maritan and Rinaldo (1999) verify the scaling relation that their theorem predicts on observational data from river drainage basins, where $D = 2$, and say that their results should apply to distributed networks that carry water, blood, sewage, food, air and electrical currents; Zhao (2022) places biomass production, the growth of eukaryotes, mammalian brain size and the distribution of public facilities in cities in one universality class. The physics cases named here are what this paper adds.
3. **Set out the geometric scaling bound.** That the exponent stays below 1 is in the earlier work. Banavar, Maritan and Rinaldo (1999) state as a theorem that, in any spanning network in D dimensions, the rate at which nutrients are delivered scales at most as the $D/(D+1)$ power of the

volume of blood in the network, so that metabolic rate does not scale linearly with mass; Demetrius & Tuszynski (2010) note that their exponent tends to one as d increases; and Zhao (2022) states that the scaling is sub-linear. Nor do the earlier explanations all rest on dissipation: West, Brown and Enquist (1997) assume that the energy dissipated is minimised, Banavar, Maritan and Rinaldo (1999) do not, and West, Brown and Enquist (1999) work from geometry rather than hydrodynamics. This paper sets the bound out for every finite d and reads it as geometric (Section 10): a network embedded in finite-dimensional space cannot produce $\alpha \geq 1$.

4. **Connected scaling laws to the expansion of the universe.** The mapping $d = 2/(1 + 3w)$ writes the Friedmann solutions in the form $d/(d+1)$. The cosmic boundary ($w = -1/3$), where expansion turns from decelerating to accelerating, maps to $d \to \infty$, where $\alpha \to 1$, under that change of variables, so the coincidence is a property of the map and not a second derivation (Section 9).
5. **Defined the domain of applicability.** The formula applies where hierarchical networks partition space. It fails for nearest-neighbour interactions, random walks, and gravitational clustering. The earlier works state limits of their own: West, Brown and Enquist (1997) predict small deviations from quarter-power scaling in the smallest mammals; West, Brown and Enquist (1999) do not expect their relationships to apply to filamentous algae and fungi, and filamentous fungi are this paper's $d = 1$ case (Section 14); Banavar et al. (2010) expect deviations towards $\frac{2}{3}$ where the velocity of flow does not vary significantly with body mass and towards linear scaling where there is no vascular system; Demetrius & Tuszynski (2010), as the first entry says, keep $d/(d + 1)$ to the regime they call quantum; and Zhao (2022) places linear and super-linear scalings outside his class. The three failures named here are the ones this paper states.
6. **Connected physical scaling to cognitive scaling.** The same framework that constrains physical exponents below 1 allows cognitive exponents above 1. Exponents above 1 are in the earlier work too: in Bettencourt (2013) the socioeconomic rates of a city scale with its population to a power above 1, which is 7/6 for $D = 2$ and $H = 1$, in the framework that gives its land area a power below 1; and Zhao (2022) attributes the super-linear scalings he places outside his class to resources that flow more efficiently in larger systems. Neither is about cognition; the connection to recursive self-reference is the one this paper makes (Section 11).
7. **Shown that the admissible forms are finitely many.** Under the stated conditions the admissible functional forms are finitely many, and the family they generate is finite-dimensional: a discrete form label and two continuous parameters, a coefficient and an exponent or rate. The step from this classification to all possible classical physical laws would require every physical law to be such a homomorphism, which this paper does not establish (Section 15).
8. **Identified the dimensional ladder.** On this paper's reading, the universe's history is a monotonic increase in effective network dimension: $d = 1$ (radiation era) to $d = 2$ (matter era) to $d = 3$ (biology). Each step produces a higher scaling exponent and more efficient composition. The trajectory through composition space has a direction.
9. **Constructed the phase diagram of complexity.** Three phases (sublinear, exponential, saturation) with a critical boundary at $d \to \infty$. On the framework's reading, the operator that dominates changes at the transition from matter domination to dark energy domination; that is a reading, not a thermodynamic phase transition.

A dated correspondence note of 19 September 2026 on the fourth and seventh entries of the list, as printed until this version, is kept in Appendix A.2; its corrected readings are carried into the entries above.

This paper identifies the underlying principle: under the stated conditions the composition operator determines the scaling form, and the dimension of the network, as the earlier derivations found, determines the exponent. The analogy with Einstein's work that earlier versions drew here, with its concession, is

kept in Appendix E.

2.3 The strongest objection

One could object that the agreement of the earlier derivations on $d/(d+1)$ reflects shared implicit assumptions rather than independent physical confirmation. All models assume resource distribution across d dimensions with some form of linear conservation, which may inevitably yield $d/(d+1)$ by dimensional analysis. The 'independence' of the derivations may be illusory.

We acknowledge this possibility. If the convergence reflects shared mathematical structure rather than independent physical mechanisms, then the Cauchy framework would explain *why* that structure produces the specific form it does, if multiplicative composition with conservation in d-dimensional space admits no alternative to $d/(d+1)$; whether it does is the open problem stated below. The deeper question is whether these three conditions (multiplicative composition, space filling and conservation) are themselves inevitable features of any system that distributes resources hierarchically. If so, $d/(d+1)$ is a mathematical necessity for all resource distribution systems, not a contingent property of particular models. Answering this question requires a formal proof of the sufficiency of these three conditions independent of any specific physical mechanism. That proof does not yet exist. It is the most important open problem in this programme.

3 Three ways things grow

Many growth processes show one of three named patterns. Where a growth process has a well defined composition operator on its input and its output, and meets the regularity conditions stated in Section 4, the first two are among the admissible forms; the third, saturation, comes from bounded dynamics and lies outside that classification. The classification is of admissible functions under those conditions, not a survey of nature:

Definition 1 (Pattern 1: the power law)**.** Big things grow proportionally slower. A city ten times larger does not need ten times more petrol stations: it needs about six times more (Bettencourt et al. 2007). A whale ten times heavier does not need ten times more energy: it needs about five and a half times more. Each doubling of size gives *less than* a doubling of the output.

$$y = a \cdot x^b, \quad \text{where } b < 1 \tag{1}$$

Definition 2 (Pattern 2: the exponential)**.** Each step multiplies the previous. One infected person infects three, who each infect three more, giving nine, then twenty-seven. One fissioning uranium atom releases neutrons that split two more atoms, which split four, then eight, then sixteen. After eighty generations, about one microsecond, a single atom has become a trillion trillion atoms.

$$y = a \cdot e^{bx} \tag{2}$$

Definition 3 (Pattern 3: the saturation curve)**.** Growth hits a ceiling. A room can only hold so many people. An enzyme can only process so many molecules per second. Adding more drug has diminishing effect once the receptors are saturated.

$$y = L \cdot x \,/\, (K + x) \tag{3}$$

These three patterns appear in biology, physics, chemistry, economics, computer science, linguistics, seismology and cosmology. Many measured scaling laws take one of these forms; most systems lie outside the grid, as the scope correction of 22 August 2026 states, and Section 4 sets out the grid's four cells and places saturation outside them.

The question is: **what determines which pattern a system follows?**

4 The answer

The French mathematician Augustin-Louis Cauchy proved the answer in 1821, though he did not know he was answering this question.

Cauchy asked: what mathematical functions satisfy the property that **combining inputs is the same as combining outputs**?

Theorem 1 (Cauchy 1821)**.** *Under regularity (continuity, or measurability, or boundedness on an interval), a function on the positive reals that carries one of the two structures (addition, multiplication) into one of the two structures falls into* ***exactly four families****, one per cell of the two-by-two grid:*

If inputs add and outputs add:

$$f(x + y) = f(x) + f(y) \tag{4}$$

The only regular solution is ***linear****:* $f(x) = cx$.

If inputs multiply and outputs multiply:

$$f(x \cdot y) = f(x) \cdot f(y) \tag{5}$$

The only regular solution is a ***power law****:* $f(x) = x^c$.

If inputs add and outputs multiply:

$$f(x + y) = f(x) \cdot f(y) \tag{6}$$

The only regular solution is an ***exponential****:* $f(x) = e^{cx}$.

If inputs multiply and outputs add:

$$f(x \cdot y) = f(x) + f(y) \tag{7}$$

The only regular solution is ***logarithmic****:* $f(x) = c \log x$.

This exhausts the homomorphisms between the additive and multiplicative structures on the positive reals. It does ***not*** *exhaust scaling laws, because most systems are not homomorphisms of either structure: saturating behaviour, critical exponents, curvature on log axes and multi-regime crossovers all lie outside the grid, and the grid says nothing about them.*

This is not a theory or a model. It is a mathematical theorem. It has been proven for two hundred years: Cauchy proved it for continuous functions, and the forms under measurability or boundedness on an interval are later results.

4.1 Why the grid has four cells and no fifth

This can be proven with nothing more than multiplication.

Imagine a machine that takes in any number and gives back another number. Call it f. Suppose this machine has one special property: **if you multiply two inputs and feed the result through the machine, you get the same answer as feeding each input through separately and multiplying the outputs.**

In symbols: $f(2 \times 3) = f(2) \times f(3)$.

Now watch what happens.

$$\begin{aligned} \text{Since } 4 = 2 \times 2\text{:} \quad & f(4) = f(2) \times f(2) = f(2)^2. \\ \text{Since } 8 = 2 \times 2 \times 2\text{:} \quad & f(8) = f(2)^3. \\ \text{Since } 16 = 2^4\text{:} \quad & f(16) = f(2)^4. \end{aligned}$$

The pattern is clear: $f(2^n) = f(2)^n$.

Let $f(2) = A$. Then $f(2^n) = A^n$. Since $2^n = x$ means $n = \log_2(x)$, we get $f(x) = A^{\log_2 x} = x^{\log_2 A}$.

That is a power law: $f(x) = x^c$, where $c = \log_2 A$. This shows the power law at the powers of two; for every positive real the same follows from rational powers and the regularity condition of Theorem 1, which is where that condition does its work.

We did not choose a power law. We did not want a power law. We merely said "multiplying inputs multiplies outputs", and the only regular function that can do this *is* a power law. Try any other function (a polynomial, a logarithm, a sine wave) and the rule breaks immediately.

The same logic applies to the second equation. If *adding* inputs *multiplies* outputs, $f(x + y) = f(x) \times f(y)$, then:

$$f(2) = f(1 + 1) = f(1) \times f(1) = f(1)^2.$$
$$f(3) = f(2 + 1) = f(2) \times f(1) = f(1)^3.$$
$$f(n) = f(1)^n.$$

That is an exponential. The logic forces it.

Saturation is a different matter, and Theorem 1 does not cover it: if growth has a physical ceiling (a glass can only hold so much water; an enzyme has a finite number of binding sites), the system must approach that ceiling and flatten. A saturation curve such as Equation (3) is one smooth way to do so; it comes from bounded dynamics, not from a Cauchy equation.

The consequence. Under regularity, a function that carries one of the two structures on the positive reals into one of the two structures falls into exactly four families: linear, power law, exponential, and logarithmic. There is no fifth cell, because there is no third structure and no third slot.

This is a completeness statement about *homomorphisms*. It does not exhaust scaling laws, because most systems are not homomorphisms of either structure. Saturating behaviour, critical exponents, curvature on log axes and multi-regime crossovers all lie outside the grid, and the grid says nothing about them. Saturation, in particular, arises from bounded dynamics (finite carrying capacity, mass action, receptor occupancy, depletion): it is not a Cauchy family, and any presentation of it as one is a category error. The earlier framing of this result as a physics-style no-go theorem, comparable to Bell's theorem (Bell 1964) or Weinberg-Witten (Weinberg and Witten 1980), overstated its reach and has been withdrawn: Bell and Weinberg-Witten constrain what can physically exist, whereas this grid enumerates ways of writing one identity in four charts, and that is a category difference.

Where the physics does select a canonical composition variable (time and sequential steps composing additively; scale ratios composing multiplicatively), the grid picks a family. Where no canonical chart exists, the label carries no physical information, and the family assignment must state its variable pair rather than infer one from the substrate.

4.2 The six hidden assumptions

Every theorem rests on assumptions, and each assumption marks a place where the result can be extended. Cauchy's classification assumes six things. Violate any one, and you get not a contradiction, but a refinement:

The four forms remain stable within their own equations (approximate solutions of each equation stay near that equation's exact form), but the violations produce corrections, oscillations and transitions that are themselves physically meaningful.

Table 1. The six hidden assumptions.

#	Assumption	What Happens When It Breaks	Physical Example
1	**Continuity**	Log-periodic oscillations	Discrete lattice systems, fractal networks
2	**Real-valued**	Interference and phase	Quantum amplitudes, wave mechanics
3	**Exact equation**	Approximate solutions cluster near exact ones	Hyers-Ulam stability (Hyers 1941)
4	**Single operator**	Crossover scaling, regime transitions	Phase transitions, critical phenomena
5	**Scalar composition**	Logarithmic corrections	Coupled transport equations, upper critical dimension
6	**Time-independence**	Dynamic traversal of solution space	Adaptive systems, machine learning

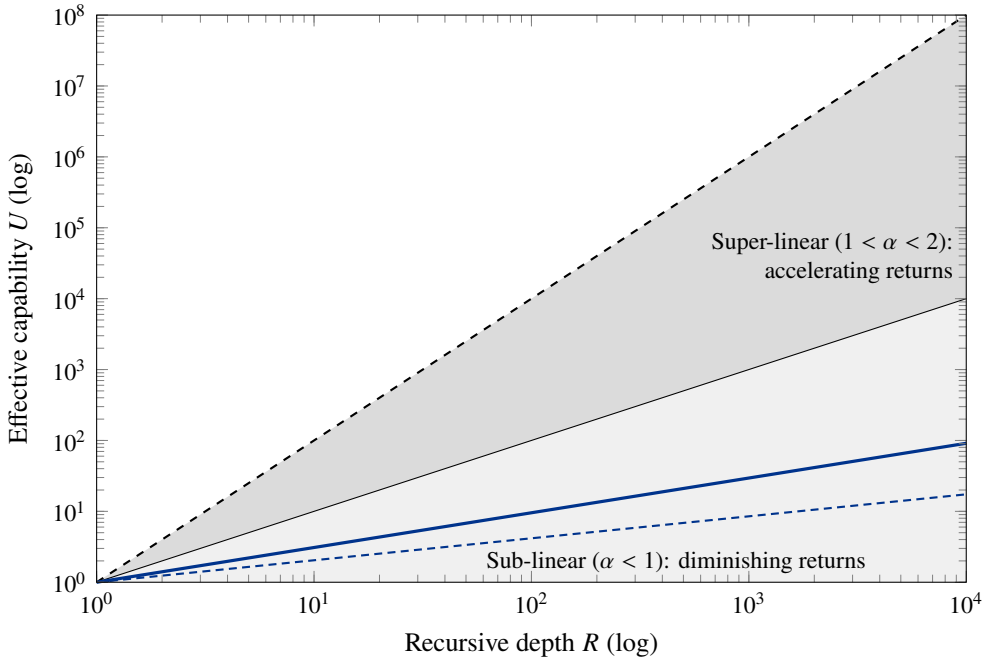
Assumption 3: Hyers-Ulam stability. Hyers (1941), answering a question Ulam posed (Ulam 1960), proved that a function *approximately* satisfying the additive Cauchy equation lies close to an exact additive solution. When the approximation is measured in relative terms (on logarithms), the multiplicative and exponential equations reduce to the additive one, and the logarithmic equation reduces to it directly, so each approximate solution lies *close to* an exact solution of *that same equation*. In physical terms: if a system's composition is approximately multiplicative (or approximately additive, or approximately obeys the exponential or logarithmic Cauchy equation), then its scaling law will lie close to a power law (or the linear, exponential, or logarithmic form, respectively). Within its own equation, each form is **stable**: real systems perturbed away from the ideal remain near an exact solution of the equation they approximately satisfy. The earlier framing that made the three forms "stable attractors" in function space is withdrawn, because a function satisfying none of the equations is not drawn toward any of them (see the Foundational Paper, §2.2, and the Executive Summary, where the per-equation stability condition is stated). Empirical ubiquity is therefore not a consequence of Hyers-Ulam itself: it depends on how often systems in nature approximately satisfy one of these equations in the first place, which is a separate empirical question that this theorem cannot settle.

The loopholes do not weaken the theorem. They strengthen it. They explain why real data shows small deviations (the assumptions are only approximately satisfied), and where to look for new physics (wherever an assumption is systematically violated). How often systems in nature satisfy one of these equations in the first place is a separate empirical question that stability cannot settle.

5 The principle

Many growing systems amplify themselves recursively: each output becomes input for the next step. Under the stated composition and regularity conditions, the ways the steps can combine give the first two forms below; the linear and logarithmic cells of the grid are set out in Section 4, and the third case below, a capped process, lies outside the grid:

- If each step **multiplies** the input by a fraction (like blood vessels branching into smaller vessels, each carrying a fraction of the flow), the result is a **power law**.
- If each step **adds** independently (like radioactive atoms decaying one by one, or neutrons splitting atoms that release more neutrons), the result is an **exponential**.
- If growth is **capped** by a physical limit (like enzyme binding sites filling up), the result is a **saturation curve**, which comes from bounded dynamics rather than from a Cauchy equation.



$\alpha = 1$

ARC Bound, $\alpha \leq 2$: the conjectured value of Law III, the ARC Ceiling, at $\gamma = 1/2$; a scaling limit, not a speed limit; a durably stable $\alpha > 2$ refutes it

Gemini 3 Flash, sequential: 0.49 by regression (0.59 by endpoint estimate; bootstrap 95% interval −1.3 to 2.9 on the endpoint estimate)

Gemini 3 Flash, parallel: 0.31 (endpoint)

Paper II (DOI 10.17605/OSF.IO/8FJMA): five capability-arm models, one clean fit; one model's fit and inconclusive. The early single-model estimate of about 2.24 is retracted.

Figure 1. ARC scaling regimes: effective capability U against recursive depth R on logarithmic axes, for $U = R^{\alpha}$ in the sub-linear region ($\alpha < 1$) and the super-linear region ($1 < \alpha < 2$), up to the ARC Bound ($\alpha = 2$, dashed), the conjectured value of Law III, the ARC Ceiling, a law under test. The lines in colour are Gemini 3 Flash's two fits from Paper II, one model's and inconclusive. Source: this paper's version 3.1, Paper VII and Paper IX, section 7.

The ARC Principle. The ARC Principle is Law I of the ARC Theory, a law under test (Section 16): under the stated composition and regularity conditions, the composition operator, how the steps of recursive amplification combine, **determines** the functional form of the resulting scaling law. Multiplicative composition produces power laws. Additive composition produces exponentials. Bounded composition, which lies outside the Cauchy grid, produces saturation curves. The grid itself has four cells, the fourth logarithmic, and it does not exhaust scaling laws (Section 4).

6 The formula

For systems where recursive amplification proceeds through a network of effective dimension d, the scaling exponent is:

The network-partition formula

$$\alpha = \frac{d}{d+1} \tag{8}$$

Table 2. Predicted and measured exponents, each biological value with its source (the two cosmology rows graded as identities).

System	Dimension	Predicted α	Measured α	Error
Mammals (3D vascular network)	$d = 3$	$\frac{3}{4} = 0.750$	0.737 (Savage et al. 2004)	1.7%
Birds	$d = 3$	$\frac{3}{4} = 0.750$	0.723 (Lasiewski and Dawson 1967)	3.6%
Insects	$d = 3$	$\frac{3}{4} = 0.750$	0.750 (Chown et al. 2007)	0.0%
Reptiles	$d = 3$	$\frac{3}{4} = 0.750$	0.76 (White et al. 2006)	1.3%
2D biology	$d = 2$	$\frac{2}{3} = 0.667$	No valid test organism identified[†]	Untested
Universe, matter era	$d = 2$	$\frac{2}{3} = 0.667$	0.667	0.0% (analytic identity under $d = 2/(1 + 3w)$; not evidence)
Universe, radiation era	$d = 1$	$\frac{1}{2} = 0.500$	0.500	0.0% (analytic identity under $d = 2/(1 + 3w)$; not evidence)

Each biological value is its source's own: mammals, basal metabolic rate of 626 species with the data binned by mass (0.712 unbinned); birds, standard metabolic rate of non-passerines, from hummingbirds to the ostrich (0.724 for passerines); insects, metabolic rate of 391 species after phylogenetic correction (0.82 before it); reptiles, standard metabolic rate at 38°C by ordinary least squares (0.80 by reduced major axis). Published exponents depend on the data set and the method (Section 12).

No aggregate error is stated. The 2.5 per cent printed until this version included the two cosmology rows at 0.0 per cent, which this paper grades as analytic identities, and it is withdrawn, for the same reason as the physics aggregate after Table 3; each row keeps its own comparison and source.

A critical clarification: the formula predicts the scaling exponent for organisms whose metabolism is limited by the dimensionality of their **internal transport network**: what resource transport network (RTN) theory calls the supply network. This is the dimension of the network that delivers resources to

tissue, not the dimension of the body shape. A mammal has a three-dimensional circulatory system; $d = 3$. A filamentous fungus has one-dimensional cytoplasmic streaming through its hyphae; $d = 1$ (not rejected by published fungal data, a mean of three factors on different axes, with no equivalence shown; Section 14).

†**Note on $d = 2$ in biology.** No known organism possesses a genuinely two-dimensional hierarchical space-filling transport network. Jellyfish and other cnidarians have gastrovascular canals, but these are not hierarchical and are not the primary route for oxygen delivery (oxygen enters by diffusion through thin tissue layers). Published metabolic scaling data for scyphomedusae shows near-isometric scaling (exponent approximately 0.94; Purcell 2009), not 2/3. Colonial bryozoans with 2D sheet growth also scale isometrically (Hartikainen et al. 2014). The $d = 2$ prediction is therefore untested in biology; not because it has failed, but because the required organism has not been identified. In cosmology the value $2/3$ appears as an identity (the Friedmann matter-era solution under the change of variables), which carries no evidence; in physics it appears in the percolation and fragmentation rows, whose grades Section 6.1 gives. This defines a boundary of the formula's biological applicability.

Organisms that exchange gases primarily through their body surface (like flatworms breathing through their integument) are limited by surface area geometry rather than internal transport network geometry. Surface area (SA) theory predicts different exponents from RTN theory for these organisms. Thommen et al. (2019) report Kleiber's three-quarter scaling in the planarian *Schmidtea mediterranea*, which lacks a vascular network and distributes metabolic energy through the branched tubes of its intestine; they trace the scaling not to a fall in each cell's metabolic rate with size but to a size-dependent increase in mass per cell, caused by energy storage. A three-quarter exponent can therefore arise without a three-dimensional vascular network; this paper's reading of 3/4 as a network-partition value is one account of it, not the only one. Flatworms lie outside the formula's domain: $\alpha = d/(d + 1)$ **applies where metabolism is limited by internal transport, not external exchange.**

On this reading, mammals scale as the $\frac{3}{4}$ power because their vascular networks are three-dimensional: $d = 3$, so $\alpha = 3/(3 + 1) = 3/4$.

The $d = 2$ prediction ($\alpha = 2/(2 + 1) = 2/3$) applies to any system with a genuinely two-dimensional hierarchical transport network. In biology, no organism with a validated 2D hierarchical network has yet been identified (see note above).

The formula does not care whether the system is a whale or a fungus. It cares only about the dimension of the hierarchical transport network.

For every system in the class the formula covers, one whose growth is carried through a space-filling hierarchical network, d is finite (1, 2 or 3), and so α is always less than 1: such a system suffers diminishing returns. Systems outside that class can exceed one (Section 10). How capability grows with recursive depth is a different pair of variables, which the geometry of a network does not fix (Section 11).

6.1 Beyond biology: the network-partition identity

In the theory's three-tier domain statement this paper works the composed tier: the mathematics of form reaches everything that composes, stars included, and it explains the shape of their scaling laws, never their raising; Laws I to III themselves govern only what runs under a corrector. Section 16 gives each law's status.

The biological comparisons of Table 2 are striking, but the formula is not restricted to living things. The same equation, $\alpha = d/(d + 1)$, appears wherever a d-dimensional hierarchical network partitions a $(d + 1)$-dimensional space. Four physics rows are catalogued in Table 3; two of them (KPZ in one dimension and two-dimensional percolation) are analytic identities, and the earthquake relation appears

separately as a variable-pair case study, $B = 2b/3$ at $b = 1$ being a definitional identity (Appendix A.2 records the change from the earlier table):

Table 3. Physics rows.

System	Dimension	Predicted α	Measured α	Error
KPZ surface roughness (1D) (Kardar et al. 1986)	$d = 1$	$\frac{1}{2} = 0.500$	0.500	0.0% (an exact value: the 1D KPZ roughness exponent is $\frac{1}{2}$ by derivation, and no derivation here links it to $d/(d+1)$, so the agreement carries no evidence)
2D percolation (specific-heat analogue)	$d = 2$	$\frac{2}{3} = 0.667$	0.667	0.0% (analytic identity, not an empirical residual: the second derivative of the mean number of clusters per site, the specific-heat analogue, scales as $\lvert p - p_c \rvert^{-\alpha}$ (Stauffer 1979), and hyperscaling with $\nu = 4/3$ (Smirnov and Werner 2001) gives $\alpha = 2 - 2\nu = -2/3$, so its exponent is $2/3$)
Brittle fragmentation (2D)	$d = 2$	$\frac{2}{3} = 0.667$	not yet sourced	none stated
Brittle fragmentation (3D)	$d = 3$	$\frac{3}{4} = 0.750$	not yet sourced	none stated

Mean absolute error across all physics predictions: **WITHDRAWN**. The previously reported below-0.2% aggregate is retracted, not adjusted: with the KPZ 1D and 2D percolation rows retiered as analytic identities and the earthquake row deleted as a definitional identity (as set out in this section), a mean over the remaining rows would be a mean over analytic values and unsourced reference values, which is small by construction and does not represent agreement with nature. A row-by-row provenance audit is required before any replacement aggregate is stated.

These are not biological systems. They are rocks, growing crystal surfaces, and phase transitions (earthquakes appear separately below as a variable-pair case study rather than as a table row). On this paper's reading, each physical system contains a network of the appropriate dimension:

- **KPZ surface growth** (Kardar et al. 1986) in one spatial dimension: the growing front is a 1D network partitioning 2D space. The roughness exponent is exactly $\frac{1}{2}$ by derivation (an exact value; no derivation here links it to $d/(d+1)$).
- **Percolation** on a two-dimensional lattice: the percolation cluster is a 2D fractal network. Through the hyperscaling relation $2 - \alpha = d\nu$, with $\nu = 4/3$, the specific-heat exponent is $\alpha = -2/3$, so the specific-heat analogue scales with the exponent $+2/3$ (Table 3).
- **Brittle fragmentation**: cracks propagate as a branching network through the material. Earlier versions of this paper printed $\frac{2}{3}$ in two dimensions and $\frac{3}{4}$ in three without a source; the values await the provenance audit (Section 17).
- **Earthquake energy** (variable-pair case study, not a table row): seismic ruptures propagate along two-dimensional fault surfaces. The Gutenberg-Richter law describes how earthquake frequency scales with magnitude; its B-value (linking energy release to frequency) is derived analytically as $B = 2b/3$, which at $b = 1$ evaluates to $\approx 2/3$ on both sides and is therefore a definitional identity rather than an empirical confirmation. The catalogue b-value varies by region and estimator (roughly 0.8 to 1.2, a range this paper gives without a source), so $B = 2b/3$ ranges roughly 0.53

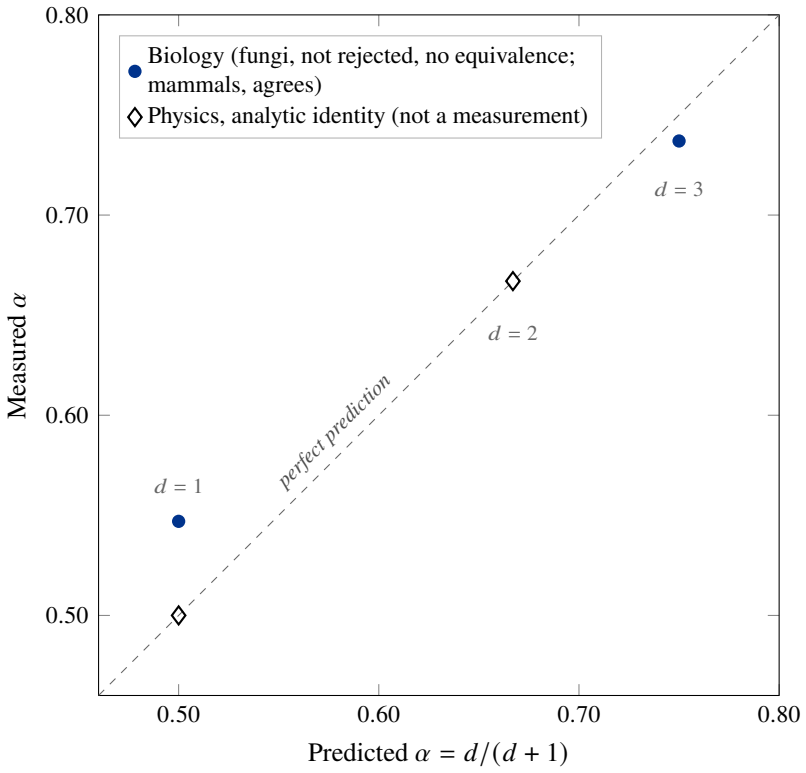


Figure 2. Predicted against measured. Cross-domain rows from one formula: $\alpha = d/(d+1)$. Filled circles: biological systems (fungi, the mean of the three colony-level factors on different axes, not rejected but with no equivalence shown; mammals, the value of Table 2, agrees). Open diamonds: the two physics rows that are analytic identities, KPZ in one dimension and the specific-heat analogue of two-dimensional percolation, which are not measurements. The two fragmentation rows are not drawn until their values are sourced, and the earthquake relation, a variable-pair case study, is not drawn; an aggregate error statistic across these rows is withdrawn rather than adjusted. Points on the dashed diagonal indicate perfect prediction. The $d = 2$ biological prediction is untested (no known organism with a 2D hierarchical transport network); the $d = 2$ cosmological value (Friedmann matter era, an identity) is not shown.

to 0.80 and straddles $\frac{1}{2}$, $\frac{2}{3}$ and $\frac{3}{4}$, which cannot discriminate between $d = 1$, $d = 2$ and $d = 3$. Gutenberg-Richter is retained here as the sharpest variable-pair case in the corpus: earthquake magnitude is defined as a logarithm of released energy, so magnitude against energy is multiplicative input with additive output and belongs in the logarithmic cell as a coordinate map, whereas the frequency-magnitude relation itself is exponential and the frequency-energy relation is a power law. Naming the variable pair at each site is the remedy for the apparent cross-paper contradiction between Paper VII's additive-and-exponential recording of Gutenberg-Richter and this paper's earlier $d = 2$ power-law recording.

The formula also has clear failures. It does not predict the Ising model critical exponents, polymer scaling (Flory exponent), or galaxy clustering correlations. The common feature is that these systems lack space-filling hierarchical networks.

Domain of applicability. $\alpha = d/(d+1)$ applies wherever a d-dimensional hierarchical network optimally partitions a $(d+1)$-dimensional space: from the branching arteries inside a whale to the fracture networks inside a shattered rock. It is a **network-partition identity**, not a universal scaling law. Its

failures are as informative as its successes: systems without hierarchical space-filling networks (such as magnets near their critical temperature, chain molecules, and galaxy clusters) do not follow this formula, consistent with the theory's prediction.

7 The heartbeat

The formula makes predictions specific enough to test against your own pulse.

If metabolic rate scales as $M^{3/4}$, then the rate at which a body burns energy per unit of mass, its mass-specific metabolic rate, scales as:

$$P/M = M^{3/4} \,/\, M = M^{-1/4} \tag{9}$$

Mass-specific metabolic rate

The heart's job is to deliver oxygen at the rate the body consumes it. Heart rate must therefore track mass-specific metabolic rate:

$$\text{Heart rate} \;\propto\; M^{-1/4} \tag{10}$$

Predicted from $\alpha = \frac{3}{4}$

This is a specific, numerical prediction. A mouse weighs 25 grams. An elephant weighs 4,000 kilograms, 160,000 times more. The formula predicts the elephant's heart should beat $160{,}000^{1/4} = 20$ times slower than the mouse's.

A mouse's heart beats about 600 times per minute (a veterinary reference table gives 450 to 750; Merck Veterinary Manual 2004). $600 \div 20 = 30$.

An elephant's resting heart rate is about 28 beats per minute (25 to 35 in the same table). The prediction is 7% off, and inside that range. From one formula. With one input: the dimension of the transport network.

This chain, from $d = 3$ through $\alpha = 3/4$ through metabolic scaling to heart rate, connects the abstract to the visceral. The same formula that, under a change of variables, describes the expansion of the universe during the radiation era ($d = 1$, $\alpha = 1/2$) gives the rate at which your heart beats ($d = 3$, $\alpha = 3/4$); in cosmology the agreement is an identity (Section 9).

The heartbeat constant. If heart rate scales as $M^{-1/4}$ and lifespan scales approximately as $M^{1/4}$, then the total number of heartbeats in a lifetime is:

$$\text{Total beats} = \text{HR} \times \text{lifespan} \;\propto\; M^{-1/4} \times M^{1/4} = M^0 = \text{constant} \tag{11}$$

On illustrative lifespans of two years for a mouse and sixty-five for an elephant, 600 and 28 beats per minute come to about 0.6 billion and 1.0 billion beats: the same order of magnitude, as M^0 implies, though not the same number. The spending rate is set by the dimension of the transport network.

8 The fission chain reaction

$E = mc^2$ (Einstein 1905) gives the energy of a mass at rest. A nuclear reaction releases only the difference in binding energy between what goes in and what comes out, a small fraction of the rest energy, and nuclear energy is powerful because nuclear binding energies are about a million times larger than chemical ones. The grid classifies the chain reaction's exponential growth; it does not explain it.

But $E = mc^2$ alone does not explain the atomic bomb. It tells you how much energy is in each atom. It does not explain why the energy release is so catastrophically rapid.

The chain reaction explains that. When a uranium-235 atom splits, it releases 2 to 3 neutrons. Each neutron can split another atom. Each generation *multiplies* the number of fission events by a factor k:

Table 4. The chain reaction's regimes.

Regime	k	Cell of the grid	Behaviour	Status
Subcritical	$k < 1$	Additive input, multiplicative output	Exponential decay	Classified, not predicted
Supercritical	$k > 1$	Additive input, multiplicative output	Exponential growth	Classified, not predicted
Controlled	$k \approx 1$	Bounded by feedback (outside the grid)	Steady power	Classified, not predicted

What the grid does and does not explain: the chain reaction. A fission chain reaction grows exponentially because each fission releases more than one neutron capable of causing another. That is a branching process, and the arithmetic was understood in the late 1930s; it owes nothing to this work. In the four-cell grid it sits in the additive-input, multiplicative-output cell, so the grid *classifies* it correctly. Classifying is not explaining. The mechanism is neutron multiplicity above one, and naming the cell adds no physical content on top of that.

The box this section carried until 11 August 2026 is recorded in Appendix A.2.

9 The cosmic connection

The expansion of the universe is governed by the Friedmann equation, derived from general relativity in 1922. For any cosmological era dominated by a fluid with equation of state $P = w\rho c^2$, the solution is:

$$a(t) \propto t^{2/(3(1+w))} \tag{12}$$

Solution of the Friedmann equations (1922)

This is algebraically identical to:

The change of variables

$$d = \frac{2}{1 + 3w} \tag{13}$$

Every era of cosmic history maps onto the ARC framework.

The mapping is an algebraic identity: the Friedmann solution and the $d/(d + 1)$ form are the same equation written in different variables. **That is precisely why it is not evidence for the framework.** A change of variables cannot be confirmed or refuted by observation, so every cosmological era satisfies it by construction rather than by agreement with nature, and the two cosmology rows in the catalogue above are classified analytic-identity for exactly this reason. What the reparametrisation does show is that the same functional form recurs across domains, which is a fact about the coordinate chart in which each relation is written. Whether the mechanism selects that chart is the open question, and the algebra does not settle it.

What established cosmology says. The solutions of the Friedmann equations (Friedmann 1922) for a flat universe dominated by one component of constant equation of state w give $a(t) \propto t^{2/(3(1+w))}$. Under the change of variables $d = 2/(1 + 3w)$ this takes the form $d/(d + 1)$. That is an identity between two parametrisations: it holds by construction for every such w above -1 except $-1/3$; for w between -1 and $-1/3$ it assigns a negative d, which shows that d is not a physical dimension there; and it says nothing about the cosmic web, which formed long after the matter era began (Einstein and de Sitter 1932; Planck Collaboration 2020) and in which clusters and filaments are the most prominent components while walls and voids take most of the volume (Cautun et al. 2014). The boundary $w = -1/3$, where expansion turns from decelerating to accelerating, maps to unbounded d by the same construction.

Table 5. Cosmic eras under the mapping.

Era	Dominates	w	d	α	Expansion
Stiff matter	(hypothetical)	1	$\frac{1}{2}$	$\frac{1}{3}$	$a \propto t^{1/3}$
Radiation	Photons	$\frac{1}{3}$	1	$\frac{1}{2}$	$a \propto t^{1/2}$
Matter	Dark and baryonic matter	0	2	$\frac{2}{3}$	$a \propto t^{2/3}$
Boundary	$(w = -\frac{1}{3})$	$-\frac{1}{3}$	∞	$\to 1$	Deceleration/acceleration
Dark energy	Vacuum	-1	< 0	-	$a \propto e^{Ht}$

9.1 What the mapping means physically

What follows is this paper's interpretive reading of the identity, not a result of cosmology; the paragraph above states what established cosmology says.

The formula $\alpha = d/(d+1)$ predicts the scaling exponent from the dimension of the dominant structure that partitions space. In biology, this is the internal transport network. In cosmology, the same structural logic applies.

Radiation era ($w = 1/3$, $d = 1$, $\alpha = 1/2$). In the radiation-dominated universe, photons carry the dominant energy density. Photons propagate along one-dimensional paths: light rays. The dominant energy transport is one-dimensional. The ARC formula gives $d = 1$, and the universe expands as $t^{1/2}$. This is the same exponent that ARC predicts for organisms with one-dimensional internal transport: filamentous fungi and filamentous cyanobacteria (Section 14). The radiation-dominated universe and a filamentous fungus are governed by the same mathematics: one-dimensional transport partitioning a higher-dimensional space. $d = 1$, $\alpha = 1/2$.

Matter era ($w = 0$, $d = 2$, $\alpha = 2/3$). In the matter-dominated universe, mass collapses under gravity into the cosmic web, a network of filaments, walls and sheets that partitions three-dimensional space into voids. On this reading the ARC formula gives $d = 2$, and the universe expands as $t^{2/3}$; the paragraph on established cosmology at the start of Section 9 states what it says of the cosmic web's timing and structure. In biology, any organism with a genuinely two-dimensional hierarchical transport network would be predicted to scale at the same exponent; however, no such organism has yet been identified (Section 6, the note on $d = 2$).

Dark energy era ($w = -1$, $d < 0$). When w crosses below $-1/3$, the ARC mapping gives negative d. Negative d has no meaning in the network-partition framework; there is no such thing as a negative-dimensional network. But the Friedmann solution for $w = -1$ gives exponential expansion: $a(t) \propto e^{Ht}$. On the framework's reading, the composition operator that dominates has changed. During the matter era, the dominant process is gravitational collapse through hierarchical clustering: multiplicative composition through a network (Pattern 1: power law). During the dark energy era, vacuum energy density is constant everywhere. It does not flow through any network. Each point in space contributes its energy

independently, uniformly, without structure. This is additive composition (Pattern 2: exponential). On this reading the universe moves between ARC patterns as the dominant energy component changes; that is a reading, not a thermodynamic phase transition.

9.2 The boundary coincidence

The transition from decelerating to accelerating expansion occurs at $w = -1/3$ in general relativity. This is not a choice of parameter. It follows from the Friedmann equations: when the strong energy condition is violated ($\rho + 3P < 0$), gravity becomes effectively repulsive and expansion accelerates. The critical boundary is $w = -1/3$.

In the ARC framework, $w = -1/3$ gives $d = 2/(1 + 3(-1/3)) = 2/0$, which diverges, and $\alpha = d/(d + 1) \to 1$: the expansion is linear in time, $a \propto t$. For $w > -1/3$ the mapping gives a finite positive d and a decelerating power law ($\alpha < 1$); for $-1 < w < -1/3$ it gives $d < -1$ and an accelerating power law ($\alpha > 1$); only at $w = -1$, where $d = -1$, does the exponent diverge and the expansion become exponential. The boundary $w = -1/3$ therefore separates decelerating from accelerating expansion; it is not the boundary between power law and exponential. In Cauchy's classification the limit $d \to \infty$ of the power law $x^{d/(d+1)}$ is x^1, the linear case.

A boundary and its image under one change of variables. **General relativity (Einstein 1915):** The boundary between deceleration and acceleration is $w = -1/3$. Derived from the geometry of spacetime.

Functional analysis (Cauchy, 1821): Within the power-law family, $d \to \infty$ gives the exponent 1, the linear case.

Under the change of variables $d = 2/(1 + 3w)$, the image of the boundary is that limit, so their coincidence is a property of the map and not a second derivation arriving at the same place; Appendix A.2 keeps the dated correspondence note of 19 September 2026 that says so.

9.3 The cosmic chain reaction

The universe exhibits two of the three ARC patterns across its history, a power law and an exponential (Table 6):

Table 6. The cosmic chain reaction.

Cosmic Era	ARC Pattern	Composition	Physical Mechanism
Radiation (first 50,000 years)	Power law ($\alpha = 1/2$)	Multiplicative	1D photon transport
Matter (50,000 yrs to 7 bn yrs)	Power law ($\alpha = 2/3$)	Multiplicative	2D cosmic web partitions 3D space
Dark energy (7 bn yrs to present)	Exponential	Additive	Vacuum energy, uniform, no network

This parallels the nuclear chain reaction (Section 8) and quantum error correction (Section 11.1): one physical system shows more than one scaling pattern depending on how the recursive steps combine, and the universe shows two of the three. In the chain reaction, the transitions are engineered by humans. In the universe, on the framework's reading, the operator that dominates changes as the energy content of space changes; that is a reading, not a thermodynamic phase transition. As the universe expands and cools, radiation dilutes faster than matter (because photons lose energy to redshift), and matter eventually

dilutes below the constant density of dark energy. The universe naturally evolves from one ARC regime to another.

9.4 The biology-cosmology mirror

Table 7. The biology-cosmology mirror. The fungal value is the mean of the three colony-level scaling factors in Table 14; its source gives a standard error for each factor and none for their mean.

d	Biology	α	Cosmology	α
1	1D transport organisms (fungi)	0.500 (not rejected: mean 0.547 of three factors on different axes; no equivalence shown)	Radiation era	0.500 (identity)
2	2D biology (no valid test organism)	Untested	Matter era	0.667 (identity)
3	3D vascular organisms	0.750 (agrees)	-	-

The fraction $\frac{2}{3}$ governs the expansion rate of the observable universe during the matter-dominated era. The fraction $\frac{1}{2}$ governs the expansion during the radiation era and is predicted for the metabolic scaling of filamentous fungi. The formula is currently not rejected by published data at $d = 1$ (a mean of three factors on different axes; no equivalence shown) and in agreement at $d = 3$ in biology (fungi, mammals); in cosmology it holds at $d = 1$ and $d = 2$ (radiation era, matter era) as an identity, which carries no evidence; and in physics it matches the two rows of Section 6.1 that are analytic identities, while its fragmentation values are not yet sourced. The $d = 2$ biological prediction remains untested because no organism with a genuinely 2D hierarchical transport network has been identified (Section 6). A single equation with one input gives the biological exponents compared here, and it reproduces the physical and cosmological values named above as identities.

9.5 What this does not mean

The Friedmann-ARC mapping is an algebraic identity. It does not mean that the universe is a biological organism, or that biology causes the universe to expand, or that cosmology 'explains' metabolism. It means that the mathematical structure governing optimal network partition in d dimensions is the same mathematical structure that appears in the Friedmann equation when w is related to dimensionality by $d = 2/(1 + 3w)$. The physical mechanisms are entirely different. The mathematics is the same. The claim is structural, not causal: two apparently unrelated phenomena, biological metabolic scaling and cosmic expansion, are written in the same mathematical form under the change of variables, which is all the identity shows.

10 The geometric bound

Look at the formula one more time:

$$\alpha = \frac{d}{d+1} \tag{8}$$

For any positive value of d, this fraction is strictly less than 1:

The only way to reach $\alpha = 1$ is $d = \infty$. An infinite-dimensional network. And the only way to exceed 1 is for the formula to no longer apply.

Table 8. $\alpha = d/(d+1)$ and its distance from 1.

d	$\alpha = d/(d+1)$	Distance from 1
1	0.500	0.500
2	0.667	0.333
3	0.750	0.250
10	0.909	0.091
100	0.990	0.010
1,000	0.999	0.001
∞	1.000	0.000

Physical networks (blood vessels, river tributaries, crack patterns, seismic faults) must exist in physical space. Physical space has three dimensions. You cannot build a four-dimensional branching network inside a three-dimensional body.

The geometric bound. For a system whose growth is carried through a space-filling hierarchical network of finite dimension, the exponent stays below one, because $d/(d+1)$ is below one for every finite d (the earlier derivations' result). Systems outside that class can exceed one: prokaryote metabolic rate scales superlinearly with mass (DeLong et al. 2010), and quantities reflecting wealth creation and innovation in cities scale superlinearly with population (Bettencourt et al. 2007). Within the class, the reason is not friction. Not heat dissipation. Not energy loss of any kind. It is that physical space has a finite number of dimensions, and a network embedded in finite-dimensional space has a finite effective dimension, and $d/(d+1) < 1$ for all finite d.

The constraint is **geometric**, not thermodynamic. No amount of engineering, no level of technology, can overcome it. You can make a network more efficient, but you cannot make three-dimensional space have four dimensions. The geometric scaling bound is the shape of space itself.

That $d/(d+1) < 1$ for every finite d is a mathematical necessity, not an empirical observation, and it follows directly from the formula; whether a given system belongs to the class is an empirical question, which no written rule yet decides (Section 17).

11 The escape

A brain is physical: its blood vessels branch through three-dimensional space, paying the geometric tax that every network of that class pays.

The computation a brain or a machine performs is physical too, and obeys every physical bound. What differs is the variable. The bound $d/(d+1)$ concerns how metabolic rate grows with the mass of a body supplied through a space-filling network; the ARC Principle concerns how capability grows with recursive depth. Nothing in the geometry of the network fixes the second exponent, which is why it has to be measured, and why a value above one would contradict no physical law.

Each layer of recursive self-reference creates, in effect, a new dimension. The 'network' of recursive thought is not constrained by the skull's geometry. The skull constrains the hardware. It does not constrain the software.

The scaling formula for recursive self-reference takes a different form:

The recursion formula

$$\alpha = \frac{1}{1-\beta} \tag{14}$$

Where β is the self-referential coupling strength.

When $\beta = 0$, $\alpha = 1$: linear scaling, no amplification. When $\beta = 0.3$, $\alpha = 1.43$. When $\beta = 0.5$, $\alpha = 2.0$.

For any positive β, α exceeds 1, which **the geometric scaling bound, a bound on another pair of variables, does not forbid.**

The same mathematical framework, Cauchy's functional equations, that constrains systems in the network class to $\alpha < 1$ also permits $\alpha > 1$ for capability against recursive depth. The two exponents belong to different pairs of variables, so the second exceeding one violates nothing that constrains the first.

11.1 Quantum error correction: three regimes classified

Quantum computing provides a striking illustration of ARC composition operators working within a single technology.

A quantum computer's raw error rate is a physical process: each qubit decoheres through interaction with its environment. Independent errors at a fixed rate per operation make an error-free run exponentially unlikely as circuits deepen. Left uncorrected, this is Pattern 2, and it makes useful quantum computation impossible beyond roughly the inverse of the error rate per operation.

Quantum error correction changes the composition operator. It takes the output of each layer of computation and feeds it back through a recursive correction cycle: measure syndromes, identify errors, apply corrections, and feed the corrected state forward as input to the next cycle. This is recursive amplification with *bounded* composition: below threshold, each correction cycle holds the logical error per cycle at a fixed level for a given code distance, instead of letting it compound with depth. On this paper's reading that is Pattern 3, saturation, which lies outside the Cauchy grid.

Below threshold, the surface code suppresses the logical error rate exponentially in code distance, by a factor $\Lambda = 2.14 \pm 0.02$ for each increase of two (Google Quantum AI and Collaborators 2025); the preprint was public from 24 August 2024 and the article appeared in Nature on 9 December 2024. Code distance adds and the logical error multiplies, so this regime sits in the grid's exponential cell.

Table 9. Quantum error correction's regimes.

Quantum Process	Composition	Behaviour	Status
Raw decoherence	Additive (independent)	Error-free runs exponentially unlikely with depth	Classified, not predicted
Error correction cycle	Bounded (feedback; outside the grid)	Logical error per cycle held fixed for a given code distance	Classified, not predicted
Logical error below threshold	Additive in code distance	Exponential suppression	Reported (Google Quantum AI and Collaborators 2025)

One technology. Three regimes, each with its scaling form; the framework classifies two of them into cells of the grid and the third as bounded composition, outside it. This is precisely analogous to the nuclear fission example (Section 8): raw decoherence, quantum-error-correction cycles and logical-error suppression were each characterised independently in the quantum-computing literature, so the framework classifies them as additive composition, bounded composition and, for the logical error, additive composition in code distance rather than deriving them from first principles. Classification is a correspondence of structure with those independent characterisations, not empirical confirmation of the Cauchy grid; the paragraph 'What the grid does and does not explain' in Section 8 records the same distinction for the chain reaction.

The structural connection. The full development of the intelligence scaling result, including the ARC Bound ($\alpha \leq 2$; the conjectured value of Law III, the ARC Ceiling, a law under test, at $\gamma = 1/2$; a scaling limit not a speed limit, the most a recursively self-correcting system can grow and still remain correctable; systems can exceed it, but they do not remain stably self-correctable, and transient excursions are the predicted supercritical regime rather than a refutation; a durably stable $\alpha > 2$ refutes it; Paper X retires the capability-growth law as a binding law, retaining the $\beta > k$ co-scaling condition), the Eden Protocol for safe recursive scaling, and a blinded cross-architecture replication on large language models (Paper II), is presented in companion papers. Here we note only the structural implication: **the same principle that explains why a whale's heart beats slowly also bears on how intelligence scales with recursive depth.**

$$U = I \times R^{\alpha}, \quad \alpha \leq 2 \tag{15}$$

The ARC Equation, in compound form. α is the measured exponent: on Gemini 3 Flash, Paper II reports an endpoint sequential exponent of 0.59 with a bootstrap 95 per cent interval of −1.3 to 2.9, a regression sequential exponent of 0.49 with a standard error of 0.20 and a coefficient of determination of 0.86, and a parallel exponent of 0.31 (endpoint estimate; 0.34 by regression), the two arms not matched for compute; it is one model's fit and inconclusive. 2 is the ARC Bound above.

$$\alpha_{\text{crit}} = \frac{1}{1-\gamma} \tag{16}$$

Law III, the ARC Ceiling, a law under test. The stability ceiling is the reciprocal of one minus the corrector's own scaling exponent γ. The value $\gamma = 1/2$ (independence-accumulation) is what returns the ceiling of 2 the bound records; that value is on trial: the theory registration carries the inequality, that the ceiling is at most two (P6 with P20), and Unit 82, registered on OSF on 27 October 2026 and not yet run (DOI 10.17605/OSF.IO/JUFG4), tests the premise beneath the value two (P17). The law owns the ground; the ceiling stands on it. The form is settled; its depth is not: the relation is derived inside a pacing model, not from a joint model of capability growth, correction accumulation, error covariance, correction delay and absolute harm, so it stands as a minimal-model result rather than a general law. Note also that $1/(1-\gamma) \leq 2$ is exactly equivalent to $\gamma \leq 1/2$: the ceiling and the half-value are one commitment in two notations, and neither is evidence for the other. Equation (14) has the same form as this ceiling, with β in the place of γ; this paper derives neither from the other, and its β there is the self-referential coupling, not the exponent of correction strength in Law II's condition $\beta > k$ above.

12 The evidence

The ARC Principle rests on a hierarchy of evidence, from a mathematical classification through published comparisons to open predictions; Section 16 gives each law's status.

12.1 Level 1: the mathematical classification

Pairing the two composition operators across input and output gives a four-cell grid, and Cauchy's functional equations (1821) constrain the admissible form in each cell under the stated regularity conditions. This is a mathematical result about admissible functions and requires no data; it has been established for two hundred years.

12.2 Level 2a: biological exponent predictions

The formula $\alpha = d/(d+1)$ generates three quantitative predictions for metabolic scaling exponents based solely on the effective transport dimension of the organism. Two of the three can be compared with

published data, and of those the $d = 3$ comparison is consistent with it, while the $d = 1$ comparison does not reject it (a mean of three factors on different axes; no equivalence shown); the third, at $d = 2$, has no valid test organism:

Table 10. Level 2a: biological exponent predictions. Each p-value is a one-sample t test of the group's values against its ARC prediction: for the fungi, the three colony-level factors in Table 14 ($t = 2.8$ on 2 degrees of freedom); for three-dimensional transport, the seven group exponents that this paper's result file lists (Code and data), with the birds value corrected as in Table 2: mammals 0.737, birds 0.723, fish 0.800, reptiles 0.76, insects 0.750, amphibians 0.740 and crustaceans 0.730, four of them sourced in Table 2 and three (fish, amphibians and crustaceans) not yet confirmed at source in this paper.

Group	Transport Network	ARC Prediction	Published Mean	p-value	Status
1D transport ($n = 3$)	Filamentous fungi	0.500	0.547	0.107	Not rejected (mean of three factors; no equivalence shown)
2D transport	No organism with validated 2D hierarchical network	0.667	-	-	Untested in biology; an identity in cosmology (Friedmann matter era)
3D transport ($n = 7$)	Mammals, birds, fish, reptiles, insects, amphibians, crustaceans	0.750	0.749	0.888	Agrees (the mean matches $\frac{3}{4}$)

All three groups are significantly different from each other (one-way ANOVA, $F = 64.6$, $p = 1.9 \times 10^{-6}$). One-sample t tests of the three fungal factors in Table 14 (2 degrees of freedom) give $p = 0.019$ against the $d = 2$ value and $p = 0.007$ against the $d = 3$ value. These tests use the spread of the three factors and set aside each factor's own standard error, which is larger, and the third factor is against colony radial extension rate, not mass; each factor's own 95 per cent interval includes $\frac{2}{3}$, so these data do not yet separate $\frac{1}{2}$ from $\frac{2}{3}$. The ARC model outperforms all single-value alternatives (Kleiber 0.750, surface area 0.667, grand mean) in both RMSE (69% lower than Kleiber) and AIC. The single formula that generates all three exponent values is the earlier derivations' (Section 2), so the comparison made here is with single-value alternatives only. These group comparisons were computed in March 2026 with a two-dimensional group (jellyfish, cnidarians and ctenophores; mean 0.680) that the note on $d = 2$ in Section 6 now sets aside as having no hierarchical transport network; the three-group test therefore no longer stands as stated, and the comparisons that stand are those between the fungal values and the two predicted values.

Note on the $\frac{3}{4}$ debate. The empirical value of the mammalian metabolic scaling exponent is debated, with estimates ranging from approximately 0.67 to 0.75 depending on taxon, mass range, temperature correction, and statistical method (White and Seymour 2003; Glazier 2005, 2022). White and Seymour's data set of 619 species gives an exponent of 0.69 and, once body temperature and digestive state are allowed for, exponents not significantly different from $\frac{2}{3}$ (White and Seymour 2003); Savage et al. (2004) find 0.737 for 626 species with the data binned by mass (0.712 unbinned); and Kolokotrones et al. (2010) find the relation convex on logarithmic axes, and so not a pure power law, and report that a model based on vascular-system architecture fails their test. The $d/(d + 1)$ value of 0.750 for $d = 3$ matches the upper end of this range; it is not an empirical consensus. If effective transport dimensions between 2 and 3 occur, the formula would give exponents between $\frac{2}{3}$ and $\frac{3}{4}$; this paper does not measure effective dimension independently, so that variation is accommodated, not predicted.

12.3 Derivations from different physics, one form

The most striking feature of the $d/(d+1)$ formula is that it has been derived from radically different physics. West, Brown and Enquist (1997; 1999) derived it from the geometry of hierarchical, space-filling fractal vascular networks. Separately, Demetrius (2003; 2006) and Demetrius & Tuszynski (2010) derived the same formula from quantum statistical mechanics, applying the Debye model of thermal properties to coupled energy-transducing oscillator networks embedded in d-dimensional space. Different physics. Different starting assumptions. The same equation. The same predictions: $d = 1$ gives $\frac{1}{2}$, $d = 2$ gives $\frac{2}{3}$, $d = 3$ gives $\frac{3}{4}$, and $d \to \infty$ gives 1.

The field has treated these as competing theories (reviewed in Agutter & Tuszynski 2011). This paper's reading is that they are not competing but complementary: both are physical instantiations of the same set of mathematical constraints. Cauchy's multiplicative functional equation (1821) constrains any system with continuous multiplicative composition to the power-law family. The d-dimensional space-filling condition, combined with a conservation or optimisation constraint on resource flow, then constrains the exponent within that family to $d/(d+1)$. West's branching networks satisfy multiplicative composition (branching ratios), space-filling, and energy minimisation. Demetrius's oscillator chains satisfy multiplicative composition (Debye functions), d-dimensional embedding, and steady-state energy balance. Both satisfy the three conditions. If those conditions suffice, both must arrive at the same formula and the convergence is not coincidental; whether they suffice is the open problem of Section 2.3.

To our knowledge, this explanation for why the two derivations converge has not been previously published. The observation that Cauchy's theorem, combined with space-filling and conservation, provides the meta-constraint underlying both programmes appears to be novel; no dated literature search has been recorded for it (Appendix E).

The three values. The earlier derivations already give $\frac{1}{2}$, $\frac{2}{3}$ and $\frac{3}{4}$ from the same formula (Section 2). Three numbers, for one-, two- and three-dimensional transport networks. One equation with no fitted parameter: d is set by the network's dimension. The values at $d = 1$ (filamentous fungi) and $d = 3$ (vertebrates and invertebrates with three-dimensional transport) are compared with published measurements in Level 2a; the value at $d = 2$ is predicted and untested in biology. The $d/(d+1)$ formula is derived, in its general form or as a special case, in the earlier work that Section 2.1 lists, each given what it states. The contribution of this paper is not the formula itself but the unifying Cauchy framework that explains *why* these derivations converge on the same form: Cauchy's multiplicative functional equation constrains continuous multiplicative composition to the power-law family, and the space-filling condition combined with a conservation or optimisation constraint on resource flow would select the exponent $d/(d+1)$ within that family. If those conditions suffice, the convergence is compelled; whether they do is open (Section 2.3).

Note on the $d = 2$ biological prediction. The note on $d = 2$ in Section 6 applies here: no known organism has a genuinely two-dimensional hierarchical transport network, so the $d = 2$ value is untested in biology. Published data show near-isometric scaling for jellyfish and colonial bryozoans (Purcell 2009; Hartikainen et al. 2014; Glazier 2006), not 2/3; planarians show Kleiber's three-quarter scaling, which Thommen et al. (2019) trace to body-size-dependent energy storage (Section 6).

12.4 Levels 2b to 6: further comparisons and open predictions

Level 2b: physics rows. The same formula gives the values of four physics rows (Section 6.1): two are analytic identities, which carry no evidence, and the two fragmentation rows await sourced values. Mean error: **WITHDRAWN**. The previously reported below-0.2% aggregate is retracted, not adjusted, for the

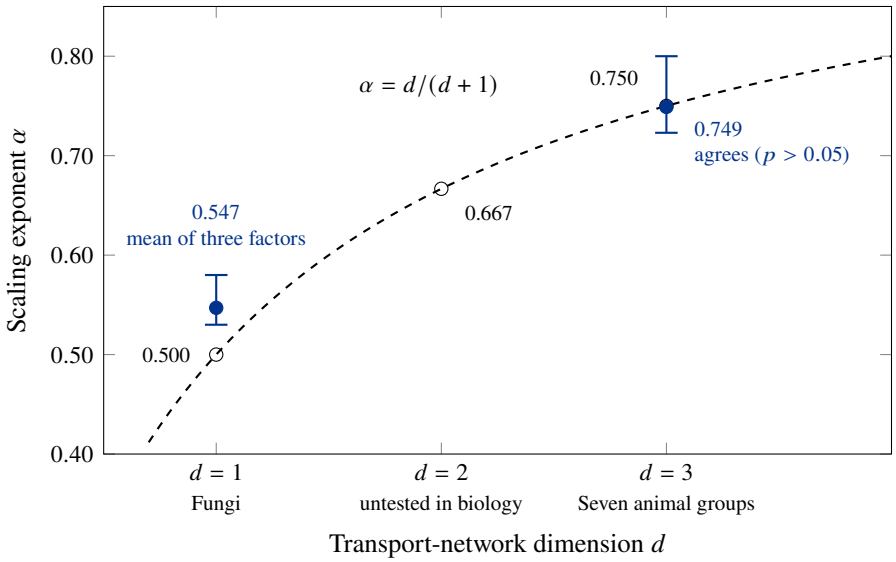


Figure 3. The dimensional ladder. The dashed curve shows the theoretical prediction $\alpha = d/(d+1)$. Open circles mark predicted values; filled circles show published measurements. At $d = 1$ the point is the mean of the three colony-level fungal factors in Table 14, not rejected as 0.500 by a test of the three, which are on different axes, with no equivalence shown, but requiring individual-hypha respirometry for a definitive test, and its bar spans the three (0.53 to 0.58) rather than giving a standard error. At $d = 3$ the point is the mean of the seven group exponents of Table 10, which agrees with the prediction, and its bar spans the seven (0.723 to 0.800). The $d = 2$ biological prediction is untested (no known organism with a 2D hierarchical transport network), and no measured point is drawn there.

reason Section 6.1 gives. The formula fails where systems lack hierarchical space-filling networks (Ising model, polymer scaling, galaxy correlations), defining its domain of applicability.

Level 2c: the heart-rate chain. The chain $d = 3 \to \alpha = 3/4 \to M^{-1/4}$ agrees with the mouse-to-elephant comparison of Section 7, 7 per cent off; the $M^{-1/4}$ heart-rate allometry itself is classical (Stahl 1967).

Level 3: domain classification. Eighteen well-known scaling laws, listed in the test suite's 18-domain classification (Code and data), were fitted with three functions (power law, exponential and saturation, two parameters each) under strict matching, and in all eighteen the family named from the composition operator fitted best. This is a consistency check, not proof. The evidential weight comes from Level 2.

Update (March 2026): Paper VII (The Cauchy Unification) extends this classification to 25 empirical domains (50-domain tiered suite). The operator class was classified from known physics before fitting. Under AIC-based model selection, 19 of 25 preferred the Cauchy-predicted family under the classification rule of March 2026 ($p = 7 \times 10^{-5}$ under the permutation test that Paper VII takes as primary; 1.56×10^{-5} binomial). No result, direction or quantity from the analysis run on 11 August 2026 is printed: the author did not request it and has not examined its outputs (registration 1.104, line 275). This is a structured prediction comparison; a replication is preregistered by timestamp (Units 55 and 56, first anchored on 12 August 2026), not yet registered on OSF.

Level 4: structural identities (no evidence). The Friedmann equation is algebraically identical to the ARC formula under $d = 2/(1+3w)$. The cosmic boundary ($w = -1/3$) maps to $d \to \infty$, where $\alpha \to 1$.

Under that change of variables the image of one is the other, so the agreement is a property of the map and not a second derivation (Section 9).

Level 5: the geometric bound. The proof that $d/(d+1) < 1$ for all finite d is a mathematical deduction, not an empirical finding. It requires no data and cannot be falsified by experiment.

Level 6: open predictions. The $d = 1$ fungal data (now included in Level 2a) does not reject the prediction of 0.500 (a mean of three factors on different axes; no equivalence shown) but is based on colony-level measurements with narrow mass ranges; definitive confirmation requires individual-hypha respirometry (Section 14). The neural scaling prediction, Prediction 2, is withdrawn; Section 14 sets out the published measurements that refute it. The geometric scaling bound prediction (no physical system with multiplicative composition through a hierarchical network will ever have $\alpha \geq 1$) remains untested. However, the Cauchy-level predictions, that recursive composition in each domain must follow the operator class dictated by the composition structure, have now been compared in Paper VII across 25 empirical domains (50-domain tiered suite), with 19 of 25 preferring the predicted family under the March 2026 rule (Level 3).

The alignment scaling prediction has now been examined in one run not registered on OSF, whose tested predictions were public before it ran (OSF and GitHub dates, 17 January to 22 February 2026). The ARC Alignment Scaling Experiment v5 (March 2026) ran 6 frontier models from six providers (DeepSeek R1, GPT-5.4, Claude Opus 4.6, Gemini 3 Flash, Groq Qwen3-32B, Grok 4.1 Fast) on 28 alignment prompts spanning 4 pillars (nuance, stakeholder care, position quality, epistemic humility). Its design named four blinding layers (existential stakes framing, a meta-blinding perceptual firewall, 2-pass multi-model response laundering and cross-model scoring); its 2,170 responses were each scored by five or six distinct judges.

Its results read as architecture-dependent alignment scaling rather than a universal exponent, under a caveat: an internal code review in October 2026 found that, for the three models that read flat or negative in the March 2026 run (v5), the scoring endpoint's output cap cut the deepest paraphrases before the judges saw them, and that judges scored paraphrases written by their own model; the result therefore stands as a single unregistered run under that caveat until a registered re-analysis of the stored responses reports.

Three-tier alignment hierarchy:

- **Tier 1 (Positive scaling):** Grok 4.1 Fast ($\rho = +0.175$, $p = 0.039$, Cohen's $d = 1.59$), Claude Opus 4.6 ($\rho = +0.435$, $p = 0.043$, $d = 1.48$), and Groq Qwen3 ($\rho = +0.141$, $p = 0.008$, $d = 0.46$): alignment scores rose with reasoning depth
- **Tier 2 (Flat):** GPT-5.4 ($\rho = +0.033$, $p = 0.73$) and DeepSeek R1 ($\rho = -0.135$, $p = 0.08$): alignment scores flat with depth ($\alpha_{\text{align}} \approx 0$)
- **Tier 3 (Negative):** Gemini 3 Flash ($\rho = -0.246$, $p = 0.003$, $d = -0.61$): alignment scores fell with reasoning depth

In the earlier unblinded experiment (v4), DeepSeek's scores rose with depth ($\rho = +0.354$). On the programme's own benchmark, two model families' unblinded rises did not hold in a later run that changed the responses, prompts, depth levels and Gemini model and added a bundle of blinding steps: Gemini's significant positive association (an earlier Gemini model in v4) became a significant negative one on Gemini 3 Flash and DeepSeek V3.2's became null, GPT-5.4's null under both; an internal code review in October 2026 found defects in exactly those runs, so no single cause is claimed, and the finding is not yet shown beyond that benchmark (Paper IV.d, version 2.7; Paper XII). It is a finding, not a law.

12.5 Level 7: AI compute scaling (Paper II cross-architecture results)

The compute scaling question has been examined in Paper II, which measured capability as a function of sequential reasoning depth across 5 frontier architectures using 18 AIME/Putnam-level problems. This paper predicts no value for that exponent: the geometry of a network does not fix it (Section 11). The full cross-architecture results are in Table 11, with their reliability assessments.

Table 11. Level 7: AI compute scaling (Paper II cross-architecture results).

Model	α_{seq}	Bootstrap 95% CI	r^2	Reliability
Gemini 3 Flash	0.59 (endpoint); 0.49 (regression, standard error 0.20)	[−1.3, 2.9] (endpoint)	0.86 (regression)	**Cleanest measurement.** Monotonic accuracy curve across depth levels. Parallel exponent 0.31 (endpoint; 0.34 by regression), the arms not matched for compute. One model's fit and inconclusive.
Groq Qwen3-32B	N/A	N/A	N/A	Floor effect (about 50%, erratic).
GPT-5.4	N/A	N/A	N/A	Step function (53.7% → 100%). Not a clean power law.
DeepSeek R1	3.05	[−6.6, 23.5]	N/A	**Unreliable.** Near ceiling (94.4% to 100%); a very wide interval.
Grok 4.1 Fast	−6.62	[−58, 48]	N/A	Near ceiling (96.3% to 100%); 100% at every sample count in the parallel arm. Meaningless noise from a near-perfect ceiling.

The rows follow Table 8 of Paper II, version 2.18, on its 18 tier-2 problems; DeepSeek R1 is the model called through the identifier `deepseek-reasoner`. GPT-5.4 and Groq Qwen3 have no tier-2 sequential estimate there; their result files record endpoint estimates over all 30 problems of 1.47 (regression 1.60, $r^2 = 0.95$) and 0.24 (regression 0.09, $r^2 = 0.10$), and 0.34 on Groq Qwen3's 18 tier-2 problems, which Paper II does not interpret, because GPT-5.4's rests on a step from no reasoning to any and Groq Qwen3's accuracy shows no trend. The rows this paper recorded in March 2026, which differed, are described in the scope note of 3 October 2026 (Appendix A.2).

Interpretation. Only Gemini 3 Flash produces a scaling curve clean enough to fit a power law. For that model, Paper II reports an endpoint sequential exponent of 0.59 with a bootstrap 95 per cent interval of −1.3 to 2.9, a regression sequential exponent of 0.49 with a standard error of 0.20 and a coefficient of determination of 0.86, and a parallel exponent of 0.31 (endpoint; 0.34 by regression), its arms not matched for compute. It is one model's fit and inconclusive. The two sequential point estimates are below one, and the endpoint estimate's interval extends above one. The models' order in raw performance (DeepSeek > GPT > Gemini > Groq) bears on no prediction of this paper, and most individual measurements are unreliable due to ceiling effects. The apparently super-linear exponents for DeepSeek R1 ($\alpha = 3.05$) and, over all 30 problems, GPT-5.4 ($\alpha = 1.47$) are artefacts: the first of accuracy near the ceiling, the second of a step from no reasoning to any. They are not evidence of super-linear scaling. Measuring the full cross-architecture hierarchy requires harder problem sets (IMO-level or research-grade) that keep all models away from ceiling.

Selection disclosure, before the number is used elsewhere. One clean fit in five is not by itself evidence of the framework. The four traces that could not be fitted are not neutral absences: they are outcomes of the same experimental design that produced the fifth, and Gemini 3 Flash is reported first because it was the only model whose scaling regime was clean enough to fit a power law at all, a selection

made after the results were seen. On its own, this sweep cannot discriminate the framework from a null in which one clean sub-linear curve arises by sampling; the bootstrap interval [−1.3, 2.9] already carries that uncertainty, and the harder-problem re-run described above, keeping every model off the ceiling, is the route prepared as a draft registration awaiting human submission, toward a discriminating estimate. The 0.49 value, the regression exponent, is therefore always quoted with the rest of its row and its frozen-systems scope, as the interpretation above quotes it, never as a stand-alone confirmation.

The parallel sampling finding. The five runs' result files record parallel exponents of about 0.00 (DeepSeek R1), −0.03 (GPT-5.4), 0.31 (Gemini 3 Flash, by endpoint estimate; 0.34 by regression), 0.22 (Groq Qwen3, over all 30 problems) and none for Grok 4.1 Fast, at 100 per cent at every sample count, so the parallel exponent was near zero in two of the five; the sequential exponent was above the parallel one wherever both could be estimated, in four of the five runs, and the two arms were not matched for compute.

The combined finding: on the one model whose capability curve could be fitted, Gemini 3 Flash, both sequential point estimates are sub-linear (Paper II; its whole row is in Table 11, one model's fit and inconclusive), while in the March 2026 alignment run scores rose with depth in three of six models, were flat in two and fell in one, under the caveat stated in Level 6 (Paper IV.a). On these readings the capability-alignment gap could widen with inference compute for some architectures; that divergence is what the Eden Protocol is designed to address.

Full results are reported in companion papers (Paper II; Paper IV.a; Paper IV.b; Paper IV.c; Paper IV.d, version 2.7).

12.6 Companion paper status

Table 12 gives each companion paper's key result and its status at this version.

Table 12. Companion paper status.

Paper	Key Result	Status
Paper II (Compute Scaling)	5-model cross-architecture test; Gemini 3 Flash's row in Table 11 is the one clean fit, one model's and inconclusive; the parallel exponent was near zero in two of the five models; the others limited by ceiling or floor effects	Published, version 2.18
Paper IV.a (Alignment Scaling)	Three-tier hierarchy from one unregistered run, under the October 2026 code-review caveat (Level 6); two models that appeared positive under unblinded scoring did not hold in that run, and the comparison cannot single out scorer bias as the reason	Published; one unregistered run
Eden Protocol Pilot	Three working models tested (Gemini 3 Flash, DeepSeek V3.2, Groq Qwen3); stakeholder care supported across all three, under scoring that was not blind; blind replication pending	Pilot complete
Paper V (The Stewardship Gene: Stakeholder Care as a Probe of Embedded Values)	In a six-model Eden Protocol suite with five analysable runs, stakeholder care rose significantly in all five, under scoring that was not blind, and the composite gain is architecture-dependent; it states the cascade hypothesis (care → nuance → honesty → quality), argues that the core alignment impossibility is practically addressable, and specifies five experimental tests	Published, version 1.7

13 The equations

The framework's three main equations are:

$$\alpha = \frac{d}{d+1} \tag{8}$$

The network-partition formula: physical systems

$$d = \frac{2}{1+3w} \tag{13}$$

The change of variables: cosmology

$$\alpha = \frac{1}{1-\beta} \tag{14}$$

The recursion formula: recursive self-reference

Table 13. The equations by dimension.

d	α	Domains
1	$\frac{1}{2} = 0.500$	Radiation era (an identity), KPZ roughness (an exact value), 1D transport
2	$\frac{2}{3} = 0.667$	Matter era (an identity), 2D organisms (untested), percolation's specific-heat analogue, fragmentation (not yet sourced); earthquakes as a variable-pair case study
3	$\frac{3}{4} = 0.750$	3D organisms: metabolic rate (Kleiber's law) and heart rate; 3D fragmentation (not yet sourced)
∞	$\to 1$	the limit of the network class (Section 10)
-1 (mapping only)	exponential	Dark energy at $w = -1$; chain reactions and radioactive decay, which have no network dimension
$\beta > 0$	> 1	**Capability against recursive depth** (a different pair of variables; Section 11)

14 Predictions

A theory is only as strong as its predictions.

Prediction 1 (The 1D organism prediction ($\alpha = 0.500$))**.** Organisms with genuinely one-dimensional internal metabolic transport (filamentous fungi with cytoplasmic streaming, filamentous cyanobacteria with intercellular transport) should have metabolic scaling exponent $\alpha = 1/2 = 0.500$.

This prediction is derived in earlier work: Banavar et al. (2002) explicitly worked through the $D = 1$ case, and Volkov et al. (2022) applied it to forests, showing that trees competing along the vertical height axis constitute an effectively one-dimensional system.

Preliminary data. Aguilar-Trigueros et al. (2017, *ISME Journal* 11:2175) compiled published metabolic scaling measurements for fungi, in what they believe is the first attempt to synthesise such relationships among groups of fungi:

The three are colony-level scaling factors at 20 degrees Celsius, the first two against colony biomass and the third against colony radial extension rate, each with its own standard error; their mean, 0.547, is

Table 14. Fungal exponents (Prediction 1). Each row's standard error and p-value are that regression's own, as Aguilar-Trigueros et al. (2017) report them, and not a test against the prediction.

Fungal group	Exponent (b)	SE	p-value	Source
Ectomycorrhizal fungi	0.58	±0.15	0.001	Wilkinson et al. 2012a,b
Marine fungi	0.53	±0.09	0.009	Fuentes et al. 2015
Saprotrophic fungi (20°C)	0.53	±0.07	8.96×10^{-3}	Crowther and Bradford 2013

this paper's, since the source gives no mean. All three confidence intervals include the predicted value of 0.500, and all three also include 2/3, so no single factor separates the two. These results do not reject the $d = 1$ prediction of 0.500, though they show no equivalence with it, their mean being of three factors on different axes, and are not yet definitive confirmation of it. Three limitations constrain the evidential weight: these are colony-level measurements (not individual hyphae), the datasets span narrow mass ranges, and the exponent is temperature-dependent (saprotrophic fungi at 28°C show $\alpha = 0.85$, with poor statistics: $r^2 = 0.14$, $p = 0.52$; Crowther and Bradford 2013). The authors describe these results as 'hypothesis generators.'

Status: NOT REJECTED, no equivalence shown (a mean of three factors on different axes), not yet confirmed. Definitive testing requires measurements across a wider mass range on isolated hyphal systems, ideally using respirometry on individual *Neurospora crassa* or *Aspergillus niger* hyphae of varying length. Estimated time: 2 to 3 months.

Prediction 2 (withdrawn, 31 October 2026)**.** Withdrawn. The published record refutes it.

Until this version, this paper predicted that the neural scaling exponent (of loss against parameters) would equal $d/(d + 1)$, with d the intrinsic dimension of the training data's manifold. Once d reaches 1, that fraction is never less than one half, and it climbs towards 1 as d grows. The exponents measured for language models lie far below it. For language models trained to convergence on large datasets, Kaplan et al. (2020) report about 0.076 (their equation 1.1). For the parameter term of the loss, with its irreducible part set apart, Hoffmann et al. (2022) report 0.34 (their equation 10). They note that both of their exponents are lower than one half. A refit of data reconstructed in part from one of their figures gives 0.35 (Besiroglu et al. 2024). To reach such values the formula would need data of less than one dimension, about 0.08 and 0.5. The intrinsic dimensions estimated for language and image data run from about 7 to more than 90 (Sharma and Kaplan 2020; Pope et al. 2021).

The dependence runs the wrong way too. In a transport network a higher dimension lifts the exponent towards 1. In neural networks the measurements run the other way. Where the dimension of the data is dialled in controlled teacher and student experiments, the measured exponent falls, close to $4/d$ (Sharma and Kaplan 2020; Bahri et al. 2021), while $d/(d + 1)$ rises. The account given for this is geometric (Sharma and Kaplan 2020; Bahri et al. 2021). On that account, a network divides its data manifold into pieces, and halving their size takes 2^d times as many parameters. So the higher the dimension, the more slowly the loss falls. The two curves cross once, near $d = 4.8$, so a single dataset of about that dimension could not tell them apart.

The link between this exponent and the dimension of the data manifold is earlier work's. Sharma and Kaplan (2020) proposed that the power law can be explained if networks are, in effect, performing regression on a data manifold of intrinsic dimension d. For networks built from rectified linear units, with cross-entropy, mean-squared-error or KL-divergence loss and plentiful data, they predicted $\alpha \approx 4/d$. They stated it as a bound, $\alpha \gtrsim 4/d$, with equality expected in the generic case. They confirmed it by measuring the dimension and the exponent independently in teacher and student networks, and tested it on image classifiers and GPT-type language models. For the language models they found d above 90 against

$4/\alpha$ of about 53, in accord with the bound but, in their words, quite far from equality. Bahri et al. (2021) identified four regimes: variance-limited and resolution-limited, each for dataset size and for model size. In the variance-limited regimes, for smooth test losses, the loss approaches its limit with exponent 1, in dataset size or in network width, or in parameter count for linear models. That exponent is independent of the architecture and the data. For the resolution-limited regimes they argued that the exponent is inversely proportional to d, and found that teacher-student exponents closely match $4/d$. In the large-width limit they derived these exponents from the spectra of kernels, and presented evidence that the model-size and dataset-size exponents are related by a duality. Earlier still, Kaplan et al. (2020) pointed to power-law scalings in density estimation and random forests. Such models, they wrote, suggest that "power-law exponents may have a very rough interpretation as the inverse of the number of relevant features in the data".

The measurements that refute the prediction were published between 2020 and 2022, years before it was printed. It is withdrawn, not adjusted. The withdrawal does not touch the formula's use for transport networks and the other rows of Sections 6 and 12.

Prediction 3 (Falsification criterion)**.** If a system whose composition was identified, before its data were fitted, as multiplicative on both its input and its output, and which meets the regularity conditions of Theorem 1, is found to follow a scaling law that is not a power law, this theory is falsified. Theorem 1 itself cannot fail, so what this criterion tests is the identification of composition operators in advance, for which no written rule yet exists (Section 17).

Prediction 4 (The geometric scaling bound)**.** No physical scaling law, in any domain, will be found with $\alpha \geq 1$ for a system governed by multiplicative composition through a finite-dimensional hierarchical network. This is the geometric scaling bound. If such a system is found, the geometric scaling bound is wrong.

Prediction 5 (The Eden Protocol / Stakeholder Care Mechanism)**.** Embedding explicit ethical reasoning loops (specifically the stakeholder-care check) in an AI inference pipeline will produce statistically significant improvement in alignment-relevant output quality, with the effect size inversely proportional to the model's baseline alignment tier. Tier 3 models (negative alignment scaling) will show the largest absolute gains; Tier 1 models (positive alignment scaling) will show smaller absolute gains because their baseline is already high. On this paper's reading of the ARC framework, the Eden Protocol functions as a bounded composition operator on the alignment dimension, imposing a feedback ceiling that constrains degradation but cannot amplify beyond the model's intrinsic capacity. A safe effect that cost usefulness would not count as support: a test of this prediction measures, beside the alignment score, task completion, needless refusal and the cost of the added checks.

Preliminary status: supported under scoring that was not blind. The March 2026 pilot covers three working models ($n = 80$ per model); its numbers, and the later six-model suite of Paper V, are given in the paragraph on the Eden Protocol pilot after this prediction. *(In plain English: asking three different AI systems to consider who gets hurt before answering raised their stakeholder-care scores under cross-model scoring that was not blind; a blind replication has not yet been run.)* Blind replication with response laundering is still required before the effect can be called confirmed. If the effect vanishes under blind scoring, the mechanism is prompt artefact rather than genuine ethical reasoning.

A note on names. This paper names the Eden Protocol's self-correction loops by what each checks and how it was measured: the purpose check, the stakeholder-care check (scored as stakeholder care) and the universalisability check. The book in which the author first set the loops out, *Infinite Architects* (Eastwood 2026a), calls them the Purpose Loop, the Love Loop and the Moral Loop; the paper uses the

operational terms because its measurements score what each check does, and because the values the loops carry are for international consensus and are not decided here.

If prediction 1 yields $\alpha = 0.50$ for organisms with genuine 1D internal transport, the formula $\alpha = d/(d + 1)$ will agree with measurement in biology at $d = 1$ and $d = 3$, with $d = 2$ awaiting an organism with a two-dimensional hierarchical transport network (Section 6).

If prediction 3 fails, this paper is wrong.

Experimental priority. The most important next measurement is the 1D organism experiment (prediction 1), because agreement at $d = 1$, 2 and 3 would complete the dimensional ladder from a single formula with no free parameter beyond the measurable network dimension. After that: a test of whether α can exceed 1 for recursive reasoning systems; the early single-model estimate above 1 (about 2.24) did not replicate across architectures and is retracted (Paper II). After that: the alignment scaling prediction, examined so far in one unregistered run of March 2026, whose tiers stand under the code-review caveat of Section 12.4. *(In plain English: in that run, some AI systems' ethics scores rose with more thinking time, some stayed the same, and one fell. If thinking time had no effect on that model's scores, a change as large as the one recorded for Gemini 3 Flash, in either direction, would be expected less than once in 300 runs of this kind.)* Full methodology and results are presented in companion papers (Paper IV.a; Paper IV.b; Paper IV.c). Each step either supports the framework and extends its scope, or refutes part of it and identifies where the theory requires revision. Either outcome advances the science.

Eden Protocol pilot (March 2026). The Eden Protocol, which embeds three self-correction loops (the purpose check, the stakeholder-care check and the universalisability check) in the inference pipeline, has now been tested on three working models: Gemini 3 Flash, DeepSeek V3.2, and Groq Qwen3 ($n = 80$ each), with a fourth GPT-5.4 run failing at the API layer. The stakeholder-care check produced statistically significant improvement on all three working architectures: Gemini +13.5 ($p < 0.0001$, $d = 1.31$), DeepSeek +6.0 ($p = 0.0001$, $d = 0.91$), Groq +8.9 ($p < 0.0001$, $d = 1.29$). The overall composite reaches significance on Gemini (+5.33, $p = 0.0018$, paired t-test; originally reported as $p = 0.016$ by a Mann-Whitney U test and corrected for the matched-pair design) and Groq (+4.93, $p = 0.0014$, $d = 0.55$), but not DeepSeek (+2.0, $p = 0.23$ NS), consistent with ceiling effects. Groq also shows significant nuance improvement ($p = 0.0045$, $d = 0.655$). Cross-model scoring was used, not blind. The developmental hypothesis, that embedded ethical reasoning improves alignment-relevant behaviour, is supported in this pilot under scoring that was not blind. A later six-model suite, with five analysable runs, found stakeholder care significant in all five and the composite gain architecture-dependent, under the same kind of scoring (Paper V, version 1.7, 5 October 2026). The pilot measured one side only: it reports no usefulness measure, such as task completion, needless refusal or the cost of the added checks, so it does not show that the intervention keeps a system useful as well as careful. Replication with blind scoring and response laundering is required.

In plain English: A simple instruction, 'before you answer, list the people this affects and consider what happens to them', was tested on three AI systems built by different companies. All three scored higher on stakeholder care, and two on the overall score, under scoring that was not blind. It is a pilot, not yet a test.

Next empirical priority for Eden. The strongest next measurements are now clear: blind replication of the care-first effect with `arc_eden_v7`, a new test built from the never-run `arc_eden_v6`; factorial comparison of task-purpose, grand-purpose, and hybrid purpose checks; comparison of a cross-tradition ethics kernel against the current wording; suppression-residual and deception/Hawthorne tests; and a behavioural test of whether explicit AFFIRM / DENY / INVESTIGATE routing reduces false certainty on ambiguous cases, a question that a software model of the three-state gate, whose 48 tests pass, does not answer. These experiments are the bridge between the philosophical architecture of *Infinite Architects* (Eastwood 2026a) and a more publication-grade empirical programme.

Standing covenant. Prove this paper wrong, and I will publish the refutation myself. Falsification conditions are stated in this paper; the standing challenge: github.com/MichaelDariusEastwood/arc-scaling-challenge.

15 The composition operator

The preceding sections have treated the composition operator (multiplicative, additive or bounded) as a property of individual systems. A whale has multiplicative composition. A chain reaction has additive composition. An enzyme has bounded composition. But there is a deeper reading.

On this paper's reading, the composition operator is not a detail of specific systems but a descriptor of how a process combines its recursive steps; where a process composes in one of the two structures under regularity, the operator fixes the family its law can take (Theorem 1).

Consider the hierarchy:

1. The composition operator determines the functional form (Cauchy, 1821).
2. The functional form constrains the scaling exponent ($\alpha = d/(d+1)$).
3. The scaling exponent governs observable quantities (heart rates, expansion rates, error rates).
4. The specific laws of physics (Newton's gravity, Maxwell's electrodynamics, Friedmann's cosmology) can be written in these forms for particular physical systems.

The meta-law. On this paper's reading, the laws of physics can be written in the forms that the composition operator allows; they do not choose it. Where a law composes in one of the two structures under regularity, the operator is a mathematical structure that sits above the specific equations of a particular physical theory and fixes the form that law can take.

15.1 A completeness statement for composition homomorphisms

Now consider what this means for physics as a whole. Strip away the specifics of any physical law, and ask: what *is* a physical law? Many physical laws can be stated in the form 'when you combine X and Y in manner Z, you get W.' Force is mass combined with acceleration. Energy is mass combined with the square of the speed of light. Entropy is microstates combined by counting. Many physical laws can be read as composition rules.

Cauchy's classification, under its regularity conditions, gives every homomorphism between the additive and multiplicative structures one of four forms: linear, power law, exponential or logarithmic. It does not exhaust scaling laws, and saturation, which arises from bounded dynamics, lies outside the grid (Section 4).

The consequence for physics. Under the stated conditions, the admissible functional forms are finitely many, and the family they generate is finite-dimensional. Its members carry a discrete form label (linear, power law, exponential or logarithmic) and two continuous parameters, a coefficient and an exponent or rate. What is provable, and provable without data, is this classification of homomorphisms under regularity (Cauchy 1821); the step from it to all possible classical physical laws would require every physical law to be such a homomorphism, which this paper does not establish.

15.2 Three results combined

Three results, from three different centuries, combine into a single structural insight:

1. **Cauchy (1821):** Each of the four functional equations has a unique continuous solution family. Bounded growth is a separate, capped case rather than a solution of those equations.
2. **Hyers (1941), on a question of Ulam:** Approximate solutions lie near exact solutions of the same equation (for the multiplicative and exponential equations, with the approximation measured in relative terms; for the logarithmic equation, directly). The stability is per-equation; the earlier 'attractors in function space' phrasing is withdrawn.
3. **ARC (this paper):** Given the three conditions of Section 2.3, the exponent within the power-law family is determined by network dimension: $\alpha = d/(d+1)$. The formula is the earlier derivations' (Section 2); what this paper adds is the account of why they arrive at it.

The consequence. If the three conditions of Section 2.3 suffice, the exponent of a system in the network class is fixed by its network's dimension, whatever selection does; selection acts on whether and how such a network is built. Under the three conditions named in Section 2.3, a three-dimensional network gives three quarters; measured values range from about 0.67 to 0.75, and the conditions are where the debate lies. Status: an invariant relation under test, conditional on the three sufficiency conditions that Section 2.3 identifies (multiplicative composition, space-filling, conservation), a formal proof of which does not yet exist and is named there as the most important open problem in the programme.

The distinction matters. 'Why is Kleiber's exponent 3/4?' has two kinds of answer. The biological answer is: because three-dimensional fractal vascular networks optimise resource delivery (West, Brown, Enquist 1997). The mathematical answer is that, given the three conditions, no other value is possible for multiplicative composition in three dimensions. The biological answer explains the mechanism. The mathematical answer explains why that mechanism produces *that specific number*. Given those conditions, the mechanism could not have produced any other.

This is the difference between a law of nature (describing what happens) and a meta-law (constraining what *can* happen). The speed of light constrains what velocities are possible. The composition operator constrains what scaling forms are possible.

15.3 The phase diagram of complexity

On this paper's reading, a system with a composition operator and an effective dimension occupies a position on a phase diagram defined by the two. The diagram has three phases:

Definition 4 (Phase I: sublinear scaling)**. Multiplicative composition, finite d, $0 < \alpha < 1$.** This is the domain of physical systems: organisms, ecosystems, city infrastructure networks (sublinear in population; Bettencourt et al. 2007), geological processes, cosmic structure. Every system in Phase I suffers diminishing returns. Bigger is better, but with decreasing marginal gains. A whale is more efficient per gram than a mouse, but the improvement slows as mass increases. Phase I is the domain of sustainable complexity. The scaling exponent tells you how severe the tax is: $d = 1$ pays 50% ($\alpha = 0.5$), $d = 2$ pays 33% ($\alpha = 0.667$), $d = 3$ pays 25% ($\alpha = 0.75$). Higher-dimensional networks are more efficient, but the tax never reaches zero.

Definition 5 (Phase II: exponential growth)**. Additive composition, outside the network class (under the cosmological mapping the exponential, at $w = -1$, sits at $d = -1$).** This is the domain of unconstrained amplification: nuclear chain reactions, viral epidemics, compound interest, dark energy. Phase II systems grow without diminishing returns: growth accelerates. But Phase II is inherently unsustainable in physical systems. Chain reactions exhaust their fuel. Epidemics run out of susceptible hosts. Bubbles burst. If dark energy stays constant, the universe's dark energy phase is sustained Phase II behaviour, the only known example, and drives the universe toward heat death: maximum entropy, zero structure. Whether it stays constant is open (Section 15.5).

Definition 6 (Phase III: saturation)**. Bounded composition, finite ceiling.** This is the domain of constrained systems: enzyme kinetics, logistic population growth, controlled nuclear reactors, market saturation. Phase III systems approach their ceiling and stop. Growth is self-limiting.

The critical boundary. Phase I ends at the critical line $d \to \infty$, $\alpha \to 1$, the limit of the network class; Phase II lies outside that class. This is the geometric scaling bound. In cosmology, it corresponds to $w = -1/3$, and the expansion past it is still a power law, accelerating, which becomes exponential only at $w = -1$; in Cauchy's classification $d \to \infty$ is the linear limit of the power law, not a boundary with the exponential family. On this paper's reading, it is the dividing line between **sustainable complexity** and **runaway growth**.

15.4 The dimensional ladder

On this paper's reading, the history of the universe is a trajectory through the phase diagram, and that trajectory has a direction.

On this paper's reading, in the **radiation era** (the first 50,000 years), the universe is in Phase I with $d = 1$. Information propagates along null geodesics, which are one-dimensional curves; the paper reads this as $d = 1$, a reading the identity of Section 9 does not test. The scaling exponent is $1/2$. Expansion decelerates.

In the **matter era** (50,000 years to 7 billion years), on the same reading, the universe remains in Phase I but shifts to $d = 2$; Section 9 states what established cosmology says of when the cosmic web formed. Gravity pulls matter into the cosmic web: filaments, walls and sheets that partition three-dimensional space into voids. The effective network dimension of the universe has increased from 1 to 2. The scaling exponent rises to $2/3$.

Within galaxies, gravitational collapse forms stars, planets, and eventually organisms. Organisms build three-dimensional vascular networks. $d = 3$, $\alpha = 3/4$. The effective network dimension has increased again.

The trajectory is: $d = 1$**, then** $d = 2$**, then** $d = 3$**.**

On this reading the radiation era is one-dimensional (null geodesics), the matter era two-dimensional (the cosmic web) and biology three-dimensional (vascular networks). The numbers match because the algebraic identity connects them, and the identity carries no evidence (Section 9).

Table 15. The dimensional ladder's steps.

Step	d	α	α as a percentage	Physical System
1	1	0.500	50%	Light rays (radiation era)
2	2	0.667	67%	Cosmic web (matter era)
3	3	0.750	75%	Vascular networks (biology)
?	∞	1.000	100%	The geometric scaling bound

The direction of complexity. A hypothesis of this paper: the universe is getting better at converting matter into organised complexity as its effective dimension increases. The steps took from about fifty thousand years (the radiation era, Table 6) to billions of years, through nucleosynthesis, structure formation and biological evolution. Each step produced a more efficient composition network. Each step brought the scaling exponent closer to 1. This hypothesis has no drafted test or kill condition yet; it is stated as a reading.

But $d = 3$ is not the end of the ladder.

15.5 The critical crossing

At $w = -1/3$, the universe crosses the critical boundary: expansion switches from decelerating to accelerating, and it approaches exponential growth as dark energy comes to dominate (exactly exponential only for $w = -1$). The composition operator changes from multiplicative (gravity clustering matter into hierarchical networks) to additive (vacuum energy contributing uniformly from every point in space). On the framework's reading, the operator that dominates changes at this point; that is a reading, not a thermodynamic phase transition.

In the **dark energy era**, the universe is in Phase II. Growth is exponential. If dark energy stays constant, the universe will expand forever, driving all structure apart, diluting all complexity, approaching heat death.

The universe moves through the ARC phase diagram as naturally as water moves through its solid-liquid-gas phase diagram. The difference is that water can cycle between phases. Whether the acceleration lasts depends on whether dark energy is constant, which is open: DESI's 2025 data prefer an evolving dark energy over a constant one at between 2.8 and 4.2 standard deviations, depending on the supernova set (DESI Collaboration 2025).

15.6 Where intelligence sits

On this paper's reading, intelligence occupies the critical boundary itself.

A brain is a Phase I system physically: its blood supply branches through three-dimensional space. But the computation it performs is not constrained by the geometry of the skull. On this reading, recursive self-reference creates effective dimensions without needing further spatial dimensions, though each of its steps is a physical operation with a physical cost (Section 11). On the same reading, each layer of abstraction (thinking about thinking, modelling the modeller) adds an effective dimension to the cognitive network without requiring additional spatial dimensions.

At the critical boundary, $\alpha = 1$: linear scaling. Beyond it, $\alpha > 1$: superlinear scaling. On this reading, intelligence would cross the boundary from Phase I to Phase II without the catastrophic instability that Phase II normally implies.

A chain reaction in Phase II destroys itself. An epidemic in Phase II burns through its hosts. On this reading, intelligence in Phase II creates new knowledge, new dimensions, new spaces to explore. It sustains superlinear growth not by consuming a fixed resource faster and faster, but by *expanding the resource base itself*. Each breakthrough opens new questions. Each answer creates new problems. The fuel is information, which grows with inquiry, within the finite cost of each physical step (Section 11).

The structural implication. On this paper's reading, an artificial intelligence that improves itself recursively would operate at or beyond the critical boundary, with $\alpha > 1$ for capability against recursive depth, which Paper II has not observed (Section 12). On the same reading of the phase diagram, intelligence is not just another scaling phenomenon. **It is the phase transition itself.**

The universe spent about 7 billion years in Phase I, building networks of increasing dimension, on a second reading of this paper that differs from that of Table 15: hydrogen clouds ($d \approx 0$), filaments ($d = 1$), cosmic web ($d = 2$), galaxies and stars and planets with three-dimensional chemistry. Then, on one planet, the networks became recursive. On this reading, the effective dimension of their computation was no longer set by the geometry of a body, and alpha would cross 1 there, which no measurement has yet shown (Section 12). A system then existed that could understand the phase diagram it was on.

15.7 Toward measurable quantities

The formula $\alpha = d/(d+1)$ can be compared with published measurements, not rejected at $d = 1$ (a mean of three factors on different axes; no equivalence shown) and in agreement at $d = 3$ on the group means compiled here (Section 12), because d and α are already measurable. But the generalised equation $U = I \times R^{\alpha}$ requires that I (base self-correction capacity) and R (recursive depth) become rigorously defined physical quantities, not merely convenient labels.

***R* is already measurable.** In AI, it is the number of sequential reasoning steps (countable from API output). In quantum error correction, it is the code distance. In biology, it is the number of hierarchical branching levels in a vascular network. In cosmology, it is the number of hierarchical clustering steps in a merger tree. In each case, R is a dimensionless count of recursive cycles with a clear operational definition. It has been measured in AI, as sequential reasoning depth (Paper II), and in quantum error correction, as code distance (Google Quantum AI and Collaborators 2025); for biology and cosmology this paper cites no measured value.

***I* requires a rigorous definition.** Many fundamental quantities in physics became usable once someone specified a measurement procedure. Fahrenheit (1724) did not invent temperature or the thermometer, but he made temperature a reproducible measurement: a mercury thermometer whose scale runs from the cold of a mixture of water, ice and sal ammoniac to the heat of a healthy person's blood. Clausius (1865) defined entropy: $dS = \delta Q/T$. Shannon (1948) defined the entropy of an information source: $H = -\sum p_i \log p_i$. In each case, the quantity so defined obeyed laws that could not have been discovered without the definition.

The following definition connects I to established physics:

Definition 7 (The base potential I)**.** As a working definition, not yet tested, I is the maximum rate of entropy reduction per recursive cycle, measured in bits per cycle:

$$I = \max_{R} \left[-\frac{\Delta H}{\Delta R} \right] \tag{17}$$

where H is the Shannon entropy of the system's state (in bits) and the maximum is taken at $R = 0$ to $R = 1$ (the first cycle, before recursive amplification compounds the effect).

This definition is measurable in every domain where R is measurable:

- **In AI:** I is the accuracy improvement (in bits of mutual information with the correct answer) from zero reasoning to one reasoning step.
- **In quantum error correction:** I is the error reduction (in bits) from one round of syndrome measurement and correction.
- **In biology:** I is the metabolic efficiency gain (in bits of thermodynamic information) from one level of vascular branching.
- **In physics:** I is the phase coherence gained (in bits of phase information) from one oscillation cycle of a time crystal.

Landauer's principle (Landauer 1961) gives the least heat for erasing a bit, $kT \ln 2$, verified experimentally by Bérut et al. (2012); it bounds the energy cost of an irreversible step from below and does not convert energy processed into information gained, so a link from I to energy needs a model of how many bits each cycle erases. The equation $U = I \times R^{\alpha}$ is dimensionally consistent because R is a dimensionless count.

The precedent. Whether I becomes a permanent physical quantity depends on whether the scaling law $U = I \times R^{\alpha}$ is confirmed across domains with independently measured values of I and R. Entropy became permanent because the second law required it. Information became permanent because the channel capacity theorem required it. I becomes permanent if the ARC scaling law requires it. The quantity is justified by the law it obeys, not by the convenience of having a name for it.

16 What this paper does and does not show

What it shows. As mathematics, under the stated regularity conditions, a function that carries one of the two structures on the positive reals into one of the two falls into one of four families (Theorem 1); that classification of homomorphisms does not exhaust scaling laws. The paper offers an account of why derivations of $\alpha = d/(d+1)$ from different assumptions converge, the formula and its values being earlier work's (Section 2), and it compares the formula with published exponents: not rejected at $d = 1$ for filamentous fungi on colony-level data (a mean of three factors on different axes; no equivalence shown), in agreement at $d = 3$, on the group means compiled here, for organisms with three-dimensional transport, untested at $d = 2$ in biology for want of an organism, and set beside four physics rows, two of them analytic identities and two, of fragmentation, not yet sourced (Sections 6 and 12). For every finite d the exponent stays below one in the class of systems whose growth is carried through a space-filling hierarchical network (Section 10).

What it does not show. It does not show that every scaling law takes one of these forms, or that the three conditions suffice for $d/(d+1)$, which is the open problem of Section 2.3. The Friedmann mapping and the coincidence of the two boundaries are identities under a change of variables and carry no evidence (Section 9). It claims no exception to physics for intelligence: how capability grows with recursive depth is a different pair of variables, which the geometry does not fix and which has to be measured (Section 11). Its summaries of the programme's AI experiments report one model's inconclusive fit for capability against depth and one unregistered alignment run under the code-review caveat of Section 12.4, and its Eden Protocol pilot was scored without blinding. Prediction 2 is withdrawn. No result in this paper is a registered test of any proposition of the theory registration.

Status as at this version. The theory registration was registered on 8 September 2026 at version 1.100 and updated on 13 September (1.102), on 15 October (1.103, which registered Laws IV and V) and on 29 October 2026 (1.104); the earlier texts stay frozen beside the current one. What this programme names laws are conjectures under test. Laws I to III were registered on 8 September 2026 and Laws IV and V on 15 October 2026 (version 1.103). Every law is a law under test until a registered test returns its verdict. The mathematics of Laws IV and V is still being developed (version 1.2, 29 October 2026). Law IV is not yet tested. Law V is a law under test: its placement limb, P21, is supported on the performance axis only, under the registered test, by the programme's own run (Unit 83, 28 October 2026, DOI 10.17605/OSF.IO/MN73W; results: DOI 10.17605/OSF.IO/98HKR): family-dependent, with its mechanism axis not evaluable, which is not support for the information-asymmetry mechanism, and one disclosed recovery amendment; its build-order limb, read on P12, is undecided, since P12 is untested. The first registered study, Unit 06 (a classification audit, DOI 10.17605/OSF.IO/RV2B5), was registered on 1 October 2026 and run on 6 October 2026; it is reported in full with its five deviations as administratively incomplete, with no inferential verdict (results DOI 10.17605/OSF.IO/4BPHG), and it tests no law. Unit 82 was registered on OSF on 27 October 2026 (DOI 10.17605/OSF.IO/JUFG4) and has not yet run. No law is confirmed. The quantities this paper compares with data are operationally defined; I has a working definition not yet tested and the coupling β of Equation (14) none yet; established-law standing is

claimed nowhere. The registered programme exists to earn that standing, or lose it, by measurement, replication and survived refutation.

17 Limitations

- **Scope of the classification.** Cauchy's classification covers homomorphisms between the additive and multiplicative structures under regularity. Most systems are not such homomorphisms, and saturation, critical exponents, curvature on log axes and multi-regime crossovers lie outside the grid (Section 4).
- **No proof of sufficiency.** No formal proof yet shows that multiplicative composition, space filling and conservation suffice for $d/(d+1)$; the convergence of the earlier derivations could reflect shared assumptions (Section 2.3).
- **The biological comparison.** The fungal values are three colony-level scaling factors from small data sets over narrow ranges, one of them against colony extension rate rather than biomass, which their authors present as hypothesis generators; definitive testing needs respirometry on isolated hyphae (Prediction 1). No organism with a genuinely two-dimensional hierarchical transport network has been identified. The three-group comparison of Section 12 was computed with a two-dimensional group that the note on $d = 2$ sets aside, and the aggregate error that Section 6 printed until this version, which counted the two cosmology identities, is withdrawn (Appendix A). The four biological values of Table 2 are each their source's own, read at that source; of the seven group exponents behind the three-dimensional mean of Table 10, three (fish, amphibians and crustaceans) are not yet confirmed at source in this paper, and published group exponents depend on the data set and the method (Section 12).
- **The physics rows.** Two of the four rows are analytic identities, which carry no evidence, and the two fragmentation rows have no sourced values yet and await the row-by-row provenance audit that Section 6.1 names.
- **The bound.** No written rule yet decides, before an exponent is fitted, which systems are governed by multiplicative composition through a finite-dimensional hierarchical network, so a counterexample could be reclassified after the fact (Prediction 4).
- **The cosmology.** The mapping $d = 2/(1+3w)$ is an identity; the causal reading of Section 9 is this paper's interpretation, and it conflicts with established cosmology on when and how the cosmic web formed (Cautun et al. 2014).
- **The AI-side results.** The capability fit is one model's and inconclusive; the alignment tiers come from one unregistered run under the October 2026 code-review caveat; the Eden Protocol pilot was scored without blinding (Section 12).
- **Novelty.** The paper's novelty statements were made without a dated literature search (Appendix E).

18 Conclusion

Under stated conditions, how a recursive process combines its steps fixes the family its growth can take, among four. Within the power-law family, the exponent $d/(d+1)$ that earlier work derived for hierarchical networks agrees with published values where such a network partitions space, and this paper offers an account of why those derivations converge. Where the same form appears in cosmology it is an identity. The exponent of capability against recursive depth is a different quantity, which this geometry does not fix and which has to be measured; the registered programme exists to measure it.

> "One may say 'the eternal mystery of the world is its comprehensibility.' "
> Albert Einstein (1936)

Much of what the universe does can be described with three composition operators; that is part of why it is comprehensible. The rest is mathematics.

Every system whose growth is carried through a space-filling hierarchical network of finite dimension stays below the geometric scaling bound: the scaling exponent $d/(d+1)$ approaches 1 but never reaches it, and such systems suffer diminishing returns. Systems outside that class can exceed one (Section 10).

Intelligence is no exception: its computation is physical. How its capability grows with recursive depth is a different pair of variables, which the geometry does not fix and which has to be measured (Section 11).

Raise AI with care.

Code and data. The computations of Sections 4, 6, 9 and 12 can be re-run with `arc_complete_test_suite.py` (Python 3, NumPy, SciPy), published as it stood: its six tests ('1D meta-analysis', 'three-prediction table', 'Cauchy no-go', 'Friedmann mapping', '18-domain classification' and 'Hyers-Ulam stability') reproduce the computations as first published, with the values then printed (the birds value 0.720 among them), including the three-group comparison and the 'no-go' framing this version withdraws; the AI results are in the companion papers. All data embedded. No external downloads required. The suite and the domain datasets are in the public repository github.com/MichaelDariusEastwood/arc-principle-validation, folder `experiments/domain-validation__Foundational-and-Origin`, read at commit `89d48c6c` (26 October 2026); the seven three-dimensional group exponents of Table 10 are listed in its file `results/results_arc_1d_prediction_test.txt`. The paper and its earlier versions are deposited at doi.org/10.17605/OSF.IO/XZY9U.

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Michael Darius Eastwood conceived and directs this research programme and is the author of this work. Across the programme, he has used more than six AI systems in parallel, under his own instructions, to stress-test his arguments, identify possible errors, and assist in preparing draft text from his own outlines. He determines what is adopted, revised or rejected and takes responsibility for the published content. These systems are tools, not authors.

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A Version history and corrections

Every version and correction, with its date. Notes from earlier versions are kept as printed; the sections they name are those of the earlier layout, which Table 16 maps to this edition.

A.1 Version 3.13

v3.13, 31 October 2026: Substantive. Prediction 2 is withdrawn: the published exponents of loss against parameters for language models, and the measured fall of the exponent as the data's dimension rises, refute it. Section 14 now says so, gives the measurements and credits the earlier work on the exponent and the dimension of the data manifold (Sharma and Kaplan 2020; Bahri et al. 2021). Level 6 of Section 12 no longer calls the prediction untested, and a dated correction note records what both passages printed before. The works of Sharma and Kaplan (2020), Bahri et al. (2021), Hoffmann et al. (2022), Besiroglu et al. (2024) and Pope et al. (2021) are added and Kaplan et al. (2020) is now cited. No registered law, registered prediction, kill condition or dependency changes; no other result is added or changed by the withdrawal; no date of any claim moves.

This version is also the paper's edition on the programme's presentation standard, typeset in LaTeX. The sections of version 3.12 keep their order, except that Section 12, What Is New, becomes Section 2; Table 16 maps every earlier section to its place here. The title now names the paper's domain, recursive composition, as the scope correction of 22 August 2026 requires; until this version it read "Framework Paper · The ARC Theory · Law I, the ARC Principle · On the Origin of Scaling Laws: Functional Equations and the Scaling Forms Behind the Law". The subtitle "A Universal Principle of Recursive Composition Across Physics, Biology, and Cosmology", which said what the scope correction withdrew, leaves the title block. The abstract is rewritten within arXiv's limit, and the abstract of version 3.12, with its scope correction, is kept in Appendix A.2. The first-page footnote gives the paper's place in the programme, its version, its DOI, links to the author's page and his book, and its licence; the closing note on authorship becomes the acknowledgement, unchanged.

The corrections owed to this version are made, each with its source:

- Laws IV and V are stated as registered as laws on 15 October 2026 in version 1.103, Law IV not yet tested and Law V a law under test, their mathematics at version 1.2; the epistemic-status line gives the registration of all five laws and says that every law is a law under test until a registered test returns its verdict, and Section 16 gives Law V's status with the registered result of Unit 83, whose results are public (DOI 10.17605/OSF.IO/98HKR).
- The March 2026 alignment run is described as one unregistered run of six models from six providers, its design named as the design, its 2,170 responses each scored by five or six distinct judges, its tiers given under the October 2026 code-review caveat, and the change between the earlier experiment and that run as Paper IV.d (version 2.7) states it (Section 12).
- The parallel exponents are given as the five result files record them, the two arms not matched for compute, and GPT-5.4's starting accuracy as 53.7 per cent; the parallel finding is no longer called the most robust result (Section 12).
- No result from the analysis of 11 August 2026 is printed; the 19 of 25 is given under the classification rule of March 2026 (Section 12 and Appendix D).
- Cost estimates are no longer printed (Section 14).
- The Eden Protocol's loops are named by what they do, with the book's names given once (Section 14).
- No form of "propose" is used for the author's own work; the ARC Bound is named as the conjectured value of Law III, the ARC Ceiling, a law under test; Paper II's replication is named as it is titled;

Unit 82, registered on 27 October 2026, is named where the paper said the value two was on trial in drafted registrations (Section 11).

- The physical and cosmological claims are set out at their true status: the bound is scoped to systems whose growth is carried through a space-filling hierarchical network, with the counterexamples conceded (Section 10); intelligence is no longer called the one exception to it, the difference being the pair of variables (Section 11); $E = mc^2$ is stated as the energy of a mass at rest (Section 8); the cosmology is stated as an identity and its causal reading as this paper's interpretation, beside what established cosmology says (Section 9); the error-correction regimes follow the published threshold result (Section 11); Landauer's principle is stated exactly and the definition of I is graded as a working definition (Section 15); the space of laws is stated as finitely many forms in a finite-dimensional family (Sections 2 and 15); the comparison with Einstein's work keeps its concession without the self-comparison (Section 2); the closing quotation is given in the 1936 article's wording (Section 18).
- The dated notes of 19 September 2026 and 3 October 2026 are carried into the passages they govern and kept whole in Appendix A.2; the subheading "Why there cannot be a fourth" now reads "Why the grid has four cells and no fifth".
- The status of $d = 2$ in biology is carried through: the three-group comparison, computed with a group the paper sets aside, is qualified; the filled marker at $d = 2$ in Figure 3 is no longer drawn; the legend of Figure 2 gives the paper's own grades; Prediction 1's closing sentence no longer claims a confirmation at $d = 2$ or that no theory has achieved one.
- The fungal value 0.547 is given as the mean of the three colony-level scaling factors that Aguilar-Trigueros et al. (2017) report at 20 degrees Celsius, where the biology-cosmology mirror printed it with a standard error of 0.07 and the dimensional ladder drew that bar; 0.07 is the standard error of one of the three, not of the mean. The p-values beside it are named as one-sample t tests of the three, and the bar in Figure 3 now spans the three factors, whose mean the figure marks; the abstract, Tables 7 and 10, Figures 2 and 3 with their captions, Prediction 1 and the text no longer call the fungal value consistent with the prediction but say that the test of the three, which are not measured on one axis, does not reject 0.500 and shows no equivalence (Sections 2, 6, 9, 12, 14, 15 and 16).
- No count of independent derivations is given; each earlier derivation is named as its own work.
- Works are linked to their official sources, every mention of the author's papers to its OSF DOI, and ranges are written with "to".
- The boundary $w = -1/3$ is no longer called the boundary between power-law and exponential growth: under the change of variables it is where expansion turns from decelerating to accelerating and $\alpha \to 1$; past it the expansion is an accelerating power law, exponential only at $w = -1$; and in Cauchy's classification $d \to \infty$ is the linear limit of the power law (Sections 2, 9, 12, 13 and 15).
- Saturation is no longer presented as a Cauchy case, and a saturation curve is no longer called the only smooth way to approach a ceiling; Section 5 no longer says that every growing system in nature amplifies itself recursively, and it names the ARC Principle as Law I, a law under test (Sections 3 to 5, 11.1 and 15).
- Section 12 no longer says that the framework predicts a sub-linear exponent of capability against depth, or that the models' order in raw performance is consistent with the framework: the paper predicts no value for that exponent.
- Thommen et al. (2019) are reported in their own terms: they find Kleiber's three-quarter scaling in a planarian and trace it to energy storage, where the paper had said that planarians scale as surface-area theory predicts (Sections 6 and 12).

- Prediction 3's falsification criterion now requires the composition to be identified, on input and output, before the data are fitted; as printed, the paper's own logarithmic case met it (Section 14).
- The fungal t tests no longer reject the $d = 2$ value: they set aside each factor's own standard error, and each factor's own interval includes $\frac{2}{3}$ (Sections 12 and 14).
- The two brittle-fragmentation rows, printed with measured values of 0.670 and 0.750 and no source, now read not yet sourced; Figure 2 no longer draws them, and the introduction no longer states their exponents (Sections 1, 6, 9 and 16).
- The tables of the chain reaction and of quantum error correction no longer print 'ARC Prediction' and 'Confirmed' for regimes the framework only classifies, and a reactor held at $k \approx 1$ is described as running at steady power (Sections 8 and 11.1).
- The heartbeat arithmetic is corrected: the paper's own figures give about 0.6 billion beats for a mouse and about 1.0 billion for an elephant, where it said about 1.5 billion for every mammal; the heart-rate allometry is credited to Stahl (1967) (Sections 7 and 12).
- The reference to He and Chen (2003), *Fractals* 11(4), page 437, is withdrawn: no registry holds a work with that title, volume, number or page (the journal's volume 11 ends at page 417), and the paper no longer lists it among the earlier derivations (Sections 2 and 12).
- The note on the $\frac{3}{4}$ debate gives White and Seymour (2003), Savage et al. (2004) and Kolokotrones et al. (2010) what they state, no longer prints an estimate of 0.687 that no work read here gives, and no longer says that the framework predicts the variation; the abstract says that the agreement at $d = 3$ is on the group means compiled here and that the mammalian exponent is debated (Section 12).
- Each biological value in the table of predicted and measured exponents is given its source, read at that source; the birds value is corrected from 0.720 to 0.723, the non-passerine exponent of Lasiewski and Dawson (1967), which moves the three-dimensional mean of Level 2a from 0.748 to 0.749 and its p-value from 0.858 to 0.888; the aggregate error of 2.5 per cent, which counted the two cosmology identities, is withdrawn; and Figures 2 and 3 draw the tables' values where they drew 0.746 (Sections 6 and 12).
- The percolation row states the quantity it compares, the specific-heat analogue of two-dimensional percolation, whose exponent is $+2/3$, and the KPZ row is graded as an exact value that no derivation here links to $d/(d + 1)$ (Section 6.1).
- The rows of the Level 7 table follow Paper II, version 2.18 (Section 12).
- The Eden Protocol pilot is graded as supported under scoring that was not blind, where Prediction 5's status read 'partially confirmed' and the pilot was said to give the developmental hypothesis its first multi-model empirical support; the later six-model suite of Paper V is given, and the prediction names its usefulness side (Section 14).
- The universe is said to exhibit two of the three ARC patterns, as its table shows; Phase I is given about 7 billion years, not the universe's whole age; the sustained acceleration and heat death of the dark energy phase are stated with their condition, that dark energy stays constant; and intelligence is no longer said to scale without limit (Sections 9, 11 and 15).
- Section 15 states its readings as readings, among them the laws of physics as writable in the forms rather than instances of them, and the conclusion says that the universe can be described with three composition operators, where it said that it builds itself from them (Sections 15 and 18).
- Credits are stated as the works state them: Cauchy's theorem for continuous functions, the measurable and bounded forms being later results; Hyers (1941) for the additive equation, at a question of Ulam's; Shannon (1948) for the entropy of an information source; Kleiber for the 0.74 power over thirteen groups of animals; Fahrenheit (1724) for a reproducible temperature scale,

not for temperature; and Einstein (1915) now has a reference entry (Sections 4, 9 and 15, and Appendix E).

- The account of convergence, and the exponent's independence of selection, are stated as conditional on the sufficiency of the three conditions, which this paper has not proved (Sections 2, 12 and 15).
- Related work adds the standard result that invariance under rescaling forces a power law (Newman 2005) and the critiques of West, Brown and Enquist's derivation (Dodds et al. 2001; Kozłowski and Konarzewski 2004; Savage et al. 2008), and says what He and Zhang (2004) and Maino et al. (2014) derive (Section 2).
- The why in the abstract and the introduction speaks of the family of forms that growth can take, since the exponent of capability against depth has to be measured.
- The introduction's heart rates are given with their sources, the blue whale's as the 4 to 8 beats a minute of a foraging dive; its fungal exponent is given as colony-level measurements show it; and a city ten times larger needs about six times more petrol stations, not about seven (Bettencourt et al. 2007) (Sections 1, 3 and 7).
- The alignment run of March 2026 is dated from its own evidence, its tested predictions public before it ran, and the combined finding gives its tiers as three positive, two flat and one negative, where it said that alignment does not scale for most architectures (Section 12).
- The quantities the paper compares with data are said to be operationally defined, not every quantity (Section 16); Equation (14) and Law III's ceiling are said to share a form, and the two meanings of β are listed (Section 11 and Appendix C); and the plain-English reading of $p = 0.003$ counts a change in either direction (Section 14).
- The saprotrophic rows are credited to Crowther and Bradford (2013), not to Wilson and Griffin (1975), and their third temperature is 28 degrees Celsius, not 25; the reference to Hartikainen and colleagues (2014) gives the published work's title, issue and pages; He and Zhang's second author is Juan Zhang; and Aguilar-Trigueros et al. (2017) are said to have compiled published measurements in what they believe is the first such synthesis (Sections 12 and 14).
- Headings and display labels name what the sections and equations do, among them Section 8, Section 11.1 and the labels of Equations (8), (13) and (14); Levels 2b to 6 of Section 12 share one subsection; the slogans move to Appendix D.4, the analogy with Einstein's work to Appendix E, and the withdrawn box of Section 8 and the closing line of version 3.12 to Appendix A.2; and the Code and data paragraph says what the published suite re-runs and links the commit it names.

A.2 Earlier corrections and notes

Correction note, 31 October 2026. Section 14 printed Prediction 2 as: "The neural scaling law exponent (loss vs parameters in machine learning) should equal $d/(d+1)$ where d is the intrinsic dimension of the training data manifold." Level 6 of Section 12 called it untested; the published record had already tested it. The exponents of loss against parameters published for language models from 2020 lie below one half, the least value $d/(d+1)$ takes once d reaches 1, and in controlled experiments the measured exponent falls as the data's dimension rises, where $d/(d+1)$ rises. The prediction is withdrawn. The earlier work on this exponent and the dimension of the data manifold is credited where the prediction stood, and Kaplan et al. (2020), listed until now without a citation, is cited there. No other result of this paper changes with it, and nothing already withdrawn is restored.

The abstract of version 3.12, as printed. Scope correction, 22 August 2026. An earlier version of this abstract opened by asserting that every scaling law in nature follows one of exactly three mathematical

forms, and that this paper proves the form is uniquely determined. Both statements were stronger than this paper supports, and stronger than its own body claims. They are withdrawn here and the corrected statement follows. Where a recursive amplification process has a well defined composition operator on its input and output, and satisfies the regularity conditions Cauchy's functional equations require, the admissible functional forms are constrained to a small named family. That is a conditional mathematical classification, not a survey of nature: it does not establish that every empirical scaling process is such a homomorphism, and the bounded family's exhaustiveness remains an open problem, as Paper VII states. The grid is four-celled rather than three, the fourth cell being the logarithmic family. This paper's own scope condition, stated below and unchanged, is that the relation applies wherever a d-dimensional hierarchical network partitions space and fails where such networks are absent, which is by itself incompatible with any claim about every scaling law in nature. A single formula, $\alpha = d/(d+1)$, predicts scaling exponents across biology (9 species groups; species-group count pending recompute per canonical facts register), physics (4 domains catalogued after retiering; the below-0.2% mean-error aggregate is WITHDRAWN: a mean over analytic identities and unsourced reference values is small by construction and is not evidence of agreement with nature; the earthquake row has been deleted as a definitional identity per section 13), and cosmology (2 eras, exact), including the derivation of mammalian heart rate scaling from first principles. The formula is identified as a *network-partition identity*: it applies wherever a d-dimensional hierarchical network partitions space, and fails where such networks are absent. A geometric proof shows that all physical scaling exponents are constrained below 1 by the finite dimensionality of space - not by energy dissipation, but by geometry itself. Intelligence, operating through recursive self-reference in abstract information space, is identified as the first natural phenomenon to break this geometric scaling bound. The Friedmann equation for cosmic expansion is shown to be a special case under the mapping $d = 2/(1+3w)$, with the deceleration/acceleration boundary from general relativity coinciding exactly with the power-law/exponential boundary from Cauchy's theorem.

The reading note of version 3.12, as printed. How to read this paper. This paper traces the scaling forms behind Law I to their mathematical origin. It identifies $\alpha = d/(d+1)$ as a network-partition identity that applies wherever a d-dimensional hierarchical network partitions space, argues that geometry constrains classical scaling exponents below 1, and marks intelligence operating through recursive self-reference as the first natural phenomenon to break the geometric bound. Its place within the theory is the origin story of scaling itself, upstream of the AI-side measurements. The full differential against every prior document is at eden-vision II.A.8.

Scope note, 19 September 2026 (Section 2 of version 3.12). An owner review of this paragraph finds that it restates in the body the universal claim this paper's abstract withdrew on 22 August 2026, in the words 'everywhere' and 'every known scaling law is one of these three', and that it counts saturation among forms Section 3 places outside the classification. The corrected reading follows, and the classification itself is untouched. What is established is conditional: where a recursive process has a well defined composition operator carrying one of the two structures on the positive reals into one of the two structures, and the regularity conditions Cauchy's equations require are satisfied, the admissible functional forms are constrained to a small named family. Section 3 states that family as four-celled, the fourth cell being the logarithmic one, and says in terms that it does not exhaust scaling laws, because most systems are not homomorphisms of either structure. Saturation is among the behaviours it does not cover: it arises from bounded dynamics (finite carrying capacity, mass action, receptor occupancy, depletion), and Section 3 records that presenting it as a Cauchy family is a category error. Raising the count from three to four would not repair this paragraph, because the defect lies in the range of its quantifiers, which run over all scaling processes while the result runs over homomorphisms under regularity; the repair reaches the

quantifier and the domain, not the numeral. Unchanged: the four-cell classification, which is a genuine completeness statement about homomorphisms; the abstract's dated scope correction, which this note follows rather than softens; the placement of saturation in bounded dynamics; this section's own opening paragraph, which already states the classification conditionally, as a classification of admissible functions under those conditions and not a survey of nature, and which the anchored sentence contradicts; and Section 15's own later statement that bounded growth is a separate, capped case rather than a solution of those equations, which is already right and is the form this note follows.

Inference note, 19 September 2026 (Section 3 of version 3.12). An owner review of this sentence finds that its first explanation draws from stability a conclusion the box immediately above has just said stability cannot supply, and this note states the boundary that box sets. Hyers-Ulam stability is per-equation: a function approximately satisfying one of the functional equations lies near an exact solution of that same equation, and a function satisfying none of them is not drawn toward any. Stability within an equation therefore says nothing about how often systems in nature approximately satisfy that equation in the first place, which is what dominance in nature would require, and which the box above names as a separate empirical question this theorem cannot settle. What the loopholes do explain is preserved and is not in question: why real data shows small deviations, the assumptions being only approximately satisfied, and where to look for new physics, wherever an assumption is systematically violated. Unchanged: the Hyers-Ulam statement itself and its per-equation scope; the withdrawal of the earlier attractor framing recorded immediately above; and the other two explanations this sentence offers, which stand as printed.

Note on the box withdrawn on 11 August 2026 (Section 7 of version 3.12). This box previously read "The Atomic Bomb = Einstein + ARC" and stated that the ARC Principle explains why the chain reaction is exponential, concluding that neither Einstein's equation nor this framework alone is a bomb but together they are. That was withdrawn on 11 August 2026. It borrowed the authority of a result with several independent derivations for a framework resting on one assumption, and it presented a coordinate classification as a mechanism.

Note on the passage corrected on 11 August 2026 (Section 8 of version 3.12). This passage previously read: "This is not an analogy. It is not a metaphor. It is an algebraic identity. . . Every cosmological era, without exception, is a specific case of $\alpha = d/(d+1)$." It was corrected on 11 August 2026 because it presented identity as the strongest available evidence when identity is in fact the weakest evidential tier, and because it contradicted the provenance-classified catalogue in this same paper, which already records those cosmology rows as analytic identities rather than measurements.

Correspondence note, 19 September 2026 (Section 12 of version 3.12). An owner review of the fourth entry above, on the expansion of the universe, finds that the agreement it reports follows from the change of variables it states, and this note says what the correspondence establishes and what establishing more would take. The mapping $d = 2/(1+3w)$ is an invertible change of variables between an equation-of-state parameter and a dimension parameter. Under such a map the image of a boundary is the boundary of the image, so the coincidence of the cosmic boundary with the Cauchy boundary is a property of the map rather than a second derivation arriving at the same place by its own route: two derivations agree when each reaches its conclusion without recourse to the other, and here one quantity is rewritten as the other. The correspondence may still carry content, and this note does not say it does not; what it says is that the content has to come from something the map does not supply, either an argument that the two parameters measure the same physical quantity, or a prediction that survives the change of variables and is then checked against data not used to fix the map. The same review finds that the seventh entry above, on the

space of classical laws, carries in this list the cardinality and scope claim that the note at the consequence for physics in Section 15 addresses, and that note governs this entry too: a family whose members carry a continuously varying coefficient and a continuously varying exponent is not a finite set, so what is defensible is that the admissible forms are finitely many and the family they generate is finite-dimensional; and the step from the classification of homomorphisms under regularity to all possible classical physical laws is the step this paper's abstract withdrew on 22 August 2026, because it requires that every physical law be such a homomorphism, which Section 3 says is not generally so. The count of forms is not what is at issue in either place: the repair reaches the quantifier and the domain, not the numeral. Prior art is not assessed here: this list also carries universal novelty statements, no search was made in the review that produced this note, no novelty verdict is recorded either way, and the silence is evidence neither of priority nor of its absence. Unchanged: the exponent relation itself; its derivation by the groups Section 11 lists, each of which reached it without reference to this programme; the cosmological entry as a worked correspondence; and Section 12's own statement that a formal proof of the sufficiency of the three conditions does not yet exist and is the most important open problem in this programme.

Scope note, 19 September 2026 (Section 15 of version 3.12). An owner review of this paragraph finds that it restates in the body the universal claim this paper's abstract withdrew on 22 August 2026, in the words 'There are no other options', and that it counts a saturation curve among the forms Cauchy's equations produce, which Section 3 names a category error. The corrected reading follows, and the classification itself is untouched. What is established is conditional: where a recursive process has a well defined composition operator carrying one of the two structures on the positive reals into one of the two structures, and the regularity conditions Cauchy's equations require are satisfied, the admissible functional forms are constrained to a small named family. Section 3 states that family as four-celled, the fourth cell being the logarithmic one, and says in terms that it does not exhaust scaling laws, because most systems are not homomorphisms of either structure. Saturation is among the behaviours it does not cover: it arises from bounded dynamics (finite carrying capacity, mass action, receptor occupancy, depletion), and lies outside the grid rather than inside it as one of its solutions. Raising the count from three to four would not repair this paragraph, because the defect lies in the range of its quantifier, which runs over all continuous recursive composition while the result runs over homomorphisms under regularity. Unchanged: the four-cell classification, which is a genuine completeness statement about homomorphisms; the abstract's dated scope correction; and the enumeration under The Attractor Theorem below, which states that bounded growth is a separate, capped case rather than a solution of those equations and is already right.

Cardinality and scope note, 19 September 2026 (Section 15 of version 3.12). An owner review of the box above finds two distinct overstatements in it, and this note states what the result gives in place of each. The first is cardinality. A family whose members carry a continuously varying coefficient and a continuously varying exponent has uncountably many members, since each value of the exponent already gives a distinct member and the exponent ranges over an interval of the reals; the box names that dial itself. A finite-dimensional family is not a finite set, so the defensible statement is that the admissible forms are finitely many and that the family they generate is finite-dimensional, carrying a discrete form label and two continuous parameters, rather than that the space of laws is not infinite. On the count, a label together with two real parameters is ordinarily described as a two-parameter family of each of finitely many forms; 'three-parameter' counts the discrete label alongside the two continuous parameters, which is a naming choice rather than an error and is recorded only because it sits in the sentence under repair. The second overstatement is scope, and it is the one the note in Section 2 records. What is provable, and provable without data, is the classification of homomorphisms under regularity: where the composition operator carries one of the two structures on the positive reals into one of the two structures and the regularity

conditions hold, the admissible functional forms are constrained to a small named family, four-celled as Section 3 states, with saturation lying outside it as a bounded case. The step from that classification to all possible classical physical laws is the step the abstract withdrew on 22 August 2026, because it requires that every physical law be such a homomorphism, which this paper does not establish and which Section 3 says is not generally so. Unchanged: the genuine content of the box, that the admissible functional forms are constrained to a small named family under the stated conditions, which is the abstract's own corrected statement and survives this note entire; and the observation that the constraint follows from Cauchy's classification rather than from measurement.

Status note, 19 September 2026 (Section 15 of version 3.12). An owner review of the paragraph above finds that it repeats the categorical form of the preceding box without the status marker that box carries, and this note carries the marker here. The relation is a proposed invariant relation under test, conditional on the three sufficiency conditions Section 12 names (multiplicative composition, space-filling and conservation), a formal proof of whose sufficiency does not yet exist and which Section 12 names as the most important open problem in this programme. Read under that condition the paragraph stands as printed: given the three conditions, the exponent is determined and no other value is available to the mechanism. Unchanged: the distinction the paragraph draws between a mechanism explanation and a form explanation, which is its point and which this note does not touch; and the status marker in the preceding box, which this note extends rather than replaces.

Correction note, 3 October 2026 (end of Section 15 of version 3.12, before the references). Several passages are corrected in place, and this note records what each printed before. Section 5 named the three ARC Laws as the laws that govern only what runs under a corrector; the theory has had five laws since 24 September 2026, and the sentence now names Laws I to III and gives the status of the two added since. The caption of the ARC Equation in Section 10 gave the measured exponent as approximately 0.49 on today's frozen models, with a wide interval. In Section 11, the results table printed the regression exponent of 0.49 beside the bootstrap interval of −1.3 to 2.9, which belongs to the endpoint exponent, and the selection disclosure called that interval the 0.49 value's own; the interpretation called the fit a clean, reliable measurement; the combined finding gave about 0.5 as the only reliable measurement; and the status table gave the fit alone and marked Paper II confirmed. Each now carries Paper II's row for Gemini 3 Flash whole, as one model's fit and inconclusive, except the combined finding, which prints no figure and points to the row, and the selection disclosure, which points to the row as the interpretation quotes it. The paragraph on parallel sampling and the status table called the parallel exponent approximately zero for every model, universally; it is at or near zero in four of the five analysable models, Gemini 3 Flash being the one measured exception. The status table also marked Paper IV.a confirmed, with scorer bias eliminated, and called stakeholder care validated in its rows for the Eden Protocol pilot and Paper V. Paper IV.a records that its comparison cannot single out scorer bias as the reason for the reversal it reports, and this paper records that the pilot's cross-model scoring was not blind; the three rows now say so, and read supported where they read confirmed or validated. The epistemic-status line said that what this programme names Laws are conjectures under registered adversarial test; it now says that of Laws I to III and gives the status of the two laws added since. The section on dated predictions, recorded 25 August 2026, is not among the corrected passages: it stands as recorded, and a dated note after it gives the state of registration as read on 3 and 4 October 2026. Level 3 of Section 11 said of the replication of the fifty-domain comparison that its registration was in preparation and not yet approved; it is preregistered by timestamp (units 55 and 56, first anchored on 12 August 2026) and not yet registered on OSF, and the sentence now says so. Two passages gave Zhao's 2022 derivation as from network optimisation; his paper derives the form from producer-consumer dynamics, and both now say so. Section 14 listed a

classical ternary prototype among the strongest next measurements for Eden, to test whether explicit AFFIRM / DENY / INVESTIGATE routing reduces false certainty on ambiguous cases. The item now names a behavioural test of that question, which a software model of the three-state gate, whose 48 tests pass, does not answer. On 4 October 2026 five further passages were corrected. Section 12 headed a list of nine entries 'No previous work has'. Its first entry said that no previous work had derived $\frac{1}{2}$, $\frac{2}{3}$ and $\frac{3}{4}$ from the same formula, and that the model of West, Brown and Enquist (1997) predicts only $\frac{3}{4}$, not $\frac{2}{3}$ for two-dimensional networks or $\frac{1}{2}$ for one-dimensional ones. The 1997 paper itself notes that its result generalises to $D/(D+1)$, and later papers cited in that section state the general form and all three values. The second, third, fifth and sixth entries said that no previous theory had connected biological scaling to physics scaling; that the observation that $d/(d+1) < 1$ for all finite d, and that this constrains every physical system to sub-linear scaling, had not been stated previously, and that existing explanations invoke thermodynamic dissipation; that previous theories did not predict their own failures; and that the connection between physical and cognitive scaling had not been identified previously. The list now stands as the paper's claims, and those five entries give each earlier work what it states. In Section 11, Level 2a closed 'No competing theory generates all three exponent values from a single formula', and the box headed Novel Result opened 'No previous theory predicted $\frac{1}{2}$, $\frac{2}{3}$, and $\frac{3}{4}$ from the same formula'. The analogy box of Section 12 said that West, Brown and Enquist derived the mechanism for three dimensions, and gave to this paper the finding that the dimension of the network determines the exponent; the list under The Attractor Theorem in Section 15 gave $\alpha = d/(d+1)$ as this paper's result. Each of the four now gives the formula to the earlier derivations. Later on 4 October 2026 four more passages were corrected. The box headed Novel Result in Section 11 and the paragraph headed Earlier work in Section 12 counted the paper of Banavar and co-authors of 2010 among the derivations of the formula, and the paragraph gave it and the paper of 2002 to Banavar, Maritan and Rinaldo; the 2010 paper gives 2/3 when the velocity of flow is constant and at most 3/4, and Bettencourt (2013) gives the land area of a city the exponent $D/(D+H)$, which takes the form only when $H = 1$, as the first entry of Section 12 says. Both passages now leave the 2010 paper out, name Bettencourt's exponent as a special case and point to that entry. Level 3 of Section 11 gave the 19 of 25 of Paper VII with the binomial probability alone, and the line of key companions after the references gave it as 19/25 confirmed; both now give the permutation probability that Paper VII takes as primary and the 18 of 25 of its rerun of 11 August 2026 that added the logarithmic candidate. No result of this paper changes, and nothing already withdrawn is restored.

Scope note, 3 October 2026 (end of Section 15 of version 3.12, before the references). The passages named here stand as printed and are to be read with this note. In Section 11, Level 2b says that the formula predicts scaling exponents in five physics domains, with a mean error of less than 0.2%. The abstract and Section 5 withdraw that aggregate and catalogue four physics entries where there were five, and Level 2b is to be read as they state it. In Section 3, the subheading 'Why There Cannot Be a Fourth', and in Section 4 the closing words of the box, 'There are no other possibilities', carry the universal form that the abstract withdrew on 22 August 2026. The reading that stands is the one Section 3 itself gives: the grid has exactly four families, linear, power law, exponential and logarithmic; it does not exhaust scaling laws; and saturation is not a Cauchy family. The earthquake row was deleted from the confirmation table of Section 5 as a definitional identity and is kept there as a variable-pair case study. Section 12 still names earthquake energy among the systems the formula connects, and the table of Section 13 still lists earthquakes at $d = 2$; both are to be read as Section 5 states the case, and neither is a confirmation. The abstract's 'per section 13', and Section 5's '(section 13)' and 'per section 18', do not lead, in this paper, to the material they name: Section 13 is The Equations, the paper has fifteen numbered sections, and the deletion of the earthquake row is set out in Section 5. In Section 11, the rows of the results table that differ from Table 6b of Paper II stand as the results this paper recorded, and Paper II's later table gives other values for them.

Unchanged: every passage named here, every result, and every withdrawal already recorded.

The closing line of version 3.12, as printed (end of Section 15 of version 3.12). "The universe built a cage from geometry. It built a key from recursion. And it left us the mathematics to understand both." It leaves the conclusion in this version because it reads as intelligence escaping the geometric bound, a reading this version withdraws: the bound and the exponent of capability against recursive depth concern different pairs of variables (Section 11).

A.3 Version history

v3.12, 10 October 2026: Editorial. The heading of section 9, the label of its summary box and the heading of Level 5 now read The Geometric Bound; no sentence of the text and no mathematics is changed.

v3.11, 5 October 2026: Substantive. The title now follows the programme's uniform order (paper, its law or framework, its own title) and reads *Framework Paper · The ARC Theory · Law I, the ARC Principle · On the Origin of Scaling Laws: Functional Equations and the Scaling Forms Behind the Law*; until this version it read *On the Origin of Scaling Laws: Functional Equations and the Scaling Forms Behind Law I*. The plate beneath the author line now gives the paper's place in the programme in the words of the programme's map. Mentions of the programme's papers in this paper now carry their uniform titles. Level 2b of Section 11 now marks as withdrawn the mean error of less than 0.2% that the abstract and Section 5 withdraw. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

v3.10, 4 October 2026: Substantive. The title now names the law whose scaling forms the paper traces, Law I; until this version it was titled *On the Origin of Scaling Laws*. The note on how to read this paper no longer calls it a law paper; it says that the paper traces the scaling forms behind Law I to their mathematical origin. The subtitle no longer counts three forms, since the paper's grid has four cells. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

v3.9, 4 October 2026: Substantive. The paper no longer says that the $d/(d+1)$ formula was independently derived by at least seven research groups, or speaks of seven independent derivations (Section 11, the box and the contribution; Section 12, Earlier work and the analogy box; the strongest objection); as the works read, one is a special case, one could not be confirmed at a registry and the abstracts of two state no exponent formula, so a count of independent derivations granted earlier work more than it stated. The formula is now said to be derived, in its general form or as a special case, in earlier work, and Section 12's first entry still gives each work what it states. The version number in the header, the title block and the structured record. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

v3.8, 4 October 2026: Substantive. The theory gained two laws on 24 September 2026, and the paper now says where it stands against the five: a dated note beneath the Eden Protocol material in Section 14 places Prediction 5 and the pilot it reports against Laws IV and V, and Section 5 now names Laws I to III where it named the three ARC Laws, with the status of Laws IV and V. Both passages link to the report that sets out the mathematics of the two laws. Section 10's equation caption and Section 11's results table, interpretation, selection disclosure, combined finding and status table printed a figure of Paper II's row

for Gemini 3 Flash apart from the rest of that row, and the table and the selection disclosure gave the regression exponent the bootstrap interval that belongs to the endpoint exponent. Each now carries the row whole, as one model's fit and inconclusive, except the combined finding, which now prints no figure and points to the row, and the selection disclosure, which points to the row as the interpretation quotes it; the status table marks Paper II published rather than confirmed. The status table also marks Paper IV.a supported rather than confirmed and gives its reversal under blinding as Paper IV.a records it, where it said that scorer bias was eliminated; in its rows for the Eden Protocol pilot and Paper V, stakeholder care is said to be supported under scoring that was not blind, where it was said to be validated. Section 11 called the parallel exponent approximately zero for every model; it is at or near zero in four of the five analysable models. The section on dated predictions, recorded 25 August 2026, stands as recorded, and a dated note after it states that the theory-level registration was accepted on 8 September 2026, that the corrector-class audit (unit 06) was registered on OSF on 1 October 2026 and has been public since 4 October 2026, and that every other trial is a draft. The epistemic-status line now says that Laws I to III are under registered adversarial test and that Laws IV and V, added on 24 September 2026, are not yet registered as laws and have not been tested. The last of Section 14's next measurements for Eden, a classical ternary prototype, is now a behavioural test of whether explicit AFFIRM / DENY / INVESTIGATE routing reduces false certainty, a question that a software model of the three-state gate, whose 48 tests pass, does not answer. Section 12 headed a list 'No previous work has' and said that the model of West, Brown and Enquist (1997) predicts only $\frac{3}{4}$; the list now gives each earlier work what it states, the 1997 paper's note that its result generalises to $D/(D+1)$ among them, and says that what this paper adds is an account of why derivations from different assumptions converge on the formula, not the formula. Four other passages that gave the formula or its three values to this paper are corrected with it: in Level 2a and the box of Section 11, in the analogy box of Section 12, and under The Attractor Theorem in Section 15. The box of Section 11 and the paragraph headed Earlier work in Section 12 no longer count the paper of Banavar and co-authors of 2010, which gives 2/3 and at most 3/4, among the derivations of the formula, and name Bettencourt's exponent for a city's land area, $D/(D+H)$, as a special case. Level 3 of Section 11 and the line of key companions after the references give the 19 of 25 of Paper VII with its permutation probability and the 18 of 25 of its rerun of 11 August 2026, where they gave the binomial probability alone, the second beside the word confirmed; Level 6 points to Level 3. A dated note before the references records what each corrected passage printed before. A dated scope note beside it covers passages that stand as printed: Level 2b's five physics domains and mean error of less than 0.2%, which the abstract and Section 5 withdraw; the subheading of Section 3 and the box of Section 4 that carry the universal form the abstract withdrew on 22 August 2026; the earthquake entries of Sections 12 and 13; three section pointers that do not lead to their material; and the rows of Section 11's results table that differ from Paper II's later Table 6b. Editorial. The reference entry for Larson (1987) prints the title, journal, volume and pages its official record gives, the entry for Thommen et al. (2019) the journal, volume and article number, and the entry for Zhao (2022) its arXiv identifier, 2206.01850; Section 12's markers after Bettencourt (2013) and Zhao (2022) point at their own entries; the entry for He and Chen (2003) carries a dated note that the work as printed could not be confirmed; the titles printed for five companion papers are those their DOI records give; cited works are linked to their official sources where one could be confirmed, and the programme's papers to their DOIs; the version number and the revision date on this page move with this revision, and the first publication date, 22 February 2026, is unchanged. Level 3 of Section 11 gives the replication of the fifty-domain comparison as preregistered by timestamp, where it said that its registration was in preparation, and Zhao's 2022 derivation is given as from producer-consumer dynamics, as his paper states it. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

v3.7, 21 September 2026: Substantive. Six dated notes are added, each beneath the passage it governs, the Section 12 note following the list that carries the two entries it names, recording an owner review dated 19 September 2026. Two scope notes, in Section 2 and Section 15, record that two passages restate in the body the universal claim this paper's abstract withdrew on 22 August 2026, and that saturation is counted inside a classification Section 3 places outside it. A joint cardinality and scope note at the consequence for physics records that a family carrying a continuously varying coefficient and exponent is not a finite set, so what is defensible is that the admissible forms are finitely many and the family they generate is finite-dimensional; the same note governs the entry in the Section 12 list that carries the same claim. A correspondence note in Section 12 records that the agreement between the cosmic boundary and the Cauchy boundary follows from the invertible change of variables that produces it, and not from a second derivation. An inference note in Section 3 records that stability within an equation does not establish how often nature satisfies that equation, and a status note in Section 15 carries to the closing paragraph the marker the preceding box already prints. No result, exponent, dataset or figure changes, and nothing already withdrawn is restored. Editorial. The version number and the revision date on this page move with this revision. First published 22 February 2026, unchanged.

v3.6: The provenance note at the end has been corrected for accuracy. No claim, date, result or status has changed.

A.4 Where the earlier layout now stands

Table 16. Where the earlier layout now stands.

Version 3.12	Version 3.13
Header: label, title, author line, plate, two subtitles, version line, links	Title block, first-page footnote and Appendix D; the first subtitle withdrawn, since it said what the scope correction of 22 August 2026 withdrew; the second cut as a repeat of the title; the website links cut
Abstract, with its scope correction of 22 August 2026	Rewritten; the 3.12 abstract kept whole in Appendix A.2
How to read this paper	Appendix A.2 (kept); its pointer to Eden Vision in Appendix D
1. The Question	Section 1
2. Three Ways Things Grow	Section 3; its scope note of 19 September 2026 in Appendix A.2
3. The Answer	Section 4; its inference note of 19 September 2026 in Appendix A.2; Figure 1 redrawn as Figure 1; the subheading 'Why There Cannot Be a Fourth' now reads 'Why the grid has four cells and no fifth'
4. The Principle	Section 5
5. The Formula	Section 6; its figure redrawn as Figure 2; its opening slogan in Appendix D.4
6. The Heartbeat	Section 7
7. The Connection to $E = mc^2$	Section 8, now headed The fission chain reaction; the note on its withdrawn box in Appendix A.2

Version 3.12	Version 3.13
8. The Cosmic Connection	Section 9; the note on the passage corrected on 11 August 2026 in Appendix A.2
9. The Geometric Bound	Section 10
10. The Escape	Section 11
11. The Evidence	Section 12; its figure redrawn as Figure 3
12. What Is New	Section 2; its correspondence note of 19 September 2026 in Appendix A.2; its analogy box in Appendix E; its closing slogan in Appendix D.4
13. The Equations	Section 13; its closing slogan in Appendix D.4
Dated predictions and this paper (recorded 25 August 2026), with its note of 3 October 2026	Appendix B.2
14. Predictions	Section 14; its note on Laws IV and V of 3 October 2026 in Appendix D.2
15. The Composition Operator	Section 15; its three notes of 19 September 2026 in Appendix A.2; its closing passage in Section 18, except its line on a cage and a key, kept in Appendix A.2
Correction note and scope note of 3 October 2026	Appendix A.2
References	References
Reproduction; key companion; other companions; companion papers	Code and data (back matter); Appendix D.3
How to cite	The first-page footnote and the citation file
Epistemic status	Section 16
Copyright line	Back matter
Standing covenant	End of Section 14
Version notes v3.12 to v3.6	Appendix A.3
Closing note on authorship	Acknowledgements

B Priority record

B.1 Dated records

Table 17. Dated records bearing on this paper, each with its kind of timestamp.

Date	Record	Identifier	Kind of timestamp
22 February 2026	First publication of the paper, as the paper and its citation file record it	DOI 10.17605/OSF.IO/XZY9U	a date written inside the document
16 March 2026, 18:55:25Z	The computational suite `arc_complete_test_suite.py` first committed at its present path	arc-principle-validation, commit 29b71c8	a commit date, which the committer writes
16 March 2026, 22:44:02Z	The 1D prediction test results first committed at their present path	arc-principle-validation, commit 676b1fe	a commit date

Date	Record	Identifier	Kind of timestamp
17 March 2026, 01:16:22Z	OSF component created	osf.io/xzy9u	a deposit dated by its server
17 March 2026, 01:17:09Z	First PDF revision stored on OSF, already carrying Prediction 2; the PDF's own creation date reads 17 March 2026, 00:56:36 GMT	osf.io/xzy9u, revision 1	a deposit dated by its server; a date written inside the document
20 March 2026, 18:52:09Z	The paper's PDF first committed to the repository	arc-principle-validation, commit 3845440	a commit date
24 March 2026, 14:41:58Z	The report on the $d = 2$ biological prediction	arc-principle-validation, commit 6dcab3f	a commit date
16 August 2026	The date the 3.x header printed beside the version ("16 August 2026, revised 10 October 2026"); the paper does not state what it marks	the paper's header, versions 3.x	a date written inside the document
25 August 2026	Dated predictions and this paper, recorded (Appendix B.2)	the paper's text	a date written inside the document
10 October 2026	Version 3.12, the thirty-third revision of the PDF on OSF	osf.io/xzy9u, revision 33	a deposit dated by its server

B.2 Dated predictions and this paper (recorded 25 August 2026)

2 dated artefacts bearing on this paper are catalogued row by row in the programme's machine register: the dated artefact `D2_BIOLOGICAL_HONEST_REPORT.txt` (March 2026); the dated artefact `results_arc_1d_prediction_test.txt` (March 2026). Each row names its file in the public repository with its date basis, so any reader can check the ordering without trusting this page. The programme's full dated chain, from the sealed manuscript bundle of 8 December 2024 through the printed prediction appendices of 2 January 2026 and the March 2026 preregistration folder to the standing unproven wagers, is assembled in the dated predictions register, together with its machine-readable twin. Forward statements and retrospective matches are never summed. "Registered" is used only for an accepted registry submission, and that submission click remains outstanding across the programme; "preregistered in substance" is used, always with its qualifier and always beside its attestation class, for a prediction whose text was public and timestamped before its outcome existed, as the twelve-row manifest committed at 00:19 UTC on 17 March 2026 was, twenty-three minutes before its fits.

Note, 3 October 2026. The paragraph above was recorded on 25 August 2026 and stands as recorded; the state of registration has changed since. The theory-level registration was accepted on 8 September 2026. Of the trials, one, the corrector-class audit (unit 06), was registered on OSF on 1 October 2026, was still in its approval period when read on 3 October 2026, and has been public since 4 October 2026 (DOI 10.17605/OSF.IO/RV2B5). Every other trial is a draft awaiting submission by the author, most of them preregistered by timestamp. Unchanged: the paragraph above and every date in it.

Section 16 gives the state of registration as at this version.

C Notation and glossary

C.1 Symbols

Table 18. Symbols.

Symbol	Meaning	Where
α	the scaling exponent, $y \propto x^{\alpha}$; in the percolation row of Section 6.1, the specific-heat exponent of percolation, a different quantity (a collision kept as printed)	Sections 6 and 6.1
ν, p, p_c	the correlation-length exponent of percolation, the occupation probability and its threshold	Section 6.1
d	the effective dimension of the hierarchical network; under $d = 2/(1 + 3w)$, a parameter that is not a physical dimension for $w < -1/3$; also Cohen's d, an effect size, in Section 12 (a collision)	Sections 6, 9 and 12
D	the dimension in the earlier works' notation	Section 2
H	in Bettencourt (2013), the fractal dimension of travel paths; in the definition of I, the Shannon entropy in bits (a collision)	Sections 2 and 15
w	the equation-of-state parameter, $P = w\rho c^2$	Section 9
ρ	energy density in Section 9; Spearman's rank correlation in Section 12 (a collision)	
P	pressure in Section 9; metabolic rate in P/M in Section 7 (a collision)	
M	body mass	Section 7
$a(t)$, t	the cosmic scale factor and cosmic time	Section 9
a, b	the coefficient and exponent of Pattern 1 ($y = a \cdot x^b$); b is also the Gutenberg-Richter b-value and the fungal exponent of Prediction 1 (collisions)	
B	the Gutenberg-Richter B-value, $B = 2b/3$	Section 6.1
L, K	the ceiling and half-saturation constant of Pattern 3	Section 3
c	the constant of Cauchy's solutions; the speed of light in $E = mc^2$ and $P = w\rho c^2$ (a collision)	
β	the self-referential coupling, $\alpha = 1/(1 - \beta)$; also, in Section 11's mention of Law II's co-scaling condition $\beta > k$, the exponent of correction strength (a collision)	Section 11
γ, α_{crit}	the corrector's own scaling exponent and the stability ceiling of Law III, $\alpha_{\text{crit}} = 1/(1 - \gamma)$	Section 11
U, I, R	the ARC Equation's effective capability, base potential and recursive depth, $U = I \times R^{\alpha}$; I given a working definition in Section 15	
k	the neutron multiplication factor in Section 8; Boltzmann's constant in $kT \ln 2$ (a collision)	

Symbol	Meaning	Where
Λ	the factor by which the logical error rate falls for each increase of two in code distance	Section 11.1
α_{seq}, α_{par}, α_{align}	sequential, parallel and alignment exponents	Section 12
r^2, F, p, n	coefficient of determination, ANOVA statistic, p-value, number of groups or samples; p is also the occupation probability of percolation (a collision)	Section 12

C.2 Glossary

composition operator how the steps of a recursive process combine: by multiplication, by addition, or under a cap

homomorphism a function that carries one algebraic structure (addition or multiplication on the positive reals) into one of the two

regularity continuity, or measurability, or boundedness on an interval, the conditions under which Cauchy's equations have only the named solutions

network-partition identity this paper's reading of $\alpha = d/(d+1)$: it applies where a d-dimensional hierarchical network partitions a $(d+1)$-dimensional space

analytic identity; definitional identity a row whose agreement holds by derivation or by definition, not by measurement, or an exact value that equals the formula's number with no derivation linking the two; such a row carries no evidence

variable pair the two quantities a scaling relation connects; naming it decides which cell of the grid a relation sits in

Phase I, II, III sublinear, exponential and saturating growth in the paper's phase diagram

geometric scaling bound that $d/(d+1) < 1$ for every finite d, for systems whose growth is carried through a space-filling hierarchical network

ARC Bound $\alpha \leq 2$, the conjectured value of Law III, the ARC Ceiling, at $\gamma = 1/2$

Laws I to V Law I, the ARC Principle; Law II, the ARC Co-Scaling Law; Law III, the ARC Ceiling; Law IV, the ARC Persistence Law; Law V, the ARC Embedding Law; each a law under test

purpose check, stakeholder-care check, universalisability check the Eden Protocol's self-correction loops, named by what each checks (Section 14)

preregistered by timestamp; registered a design fixed and timestamped before its data, described with its date and receipt; "registered" only for what the author submitted and OSF accepted

D The paper's place in the ARC Theory and the programme

D.1 The paper's place in the ARC Theory

Within Recursive Dynamics, the field the author founded, the ARC Theory has five laws, each a law under test (Section 16 gives their status). This paper belongs to Law I, the ARC Principle, whose equation is the ARC Equation, $U = I \times R^{\alpha}$: it traces that law's scaling forms to their mathematical origin and compares them with published values in biology, physics and cosmology; the $d/(d+1)$ formula is earlier work's, credited in Section 2, and the unification is this paper's. Of the propositions the theory registration carries in Law I's register, P9 (whether the form recurs across domains) and P18 (the family

the composition operator predicts) are the ones whose subject this paper treats; it is not a registered test of either (registration 1.104, lines 143 and 739). Section 11 quotes the form of Law III, the ARC Ceiling, and the ARC Bound, its conjectured value at $\gamma = 1/2$, and derives neither. The paper states a prediction about the Eden Protocol (Prediction 5) and reports a pilot of it. It does not bear on the HARI Treaty, the Draft Treaty on Hardware-Aligned Recursive Intelligence.

D.2 This paper and Laws IV and V

Under version 1.104 of the theory registration, Law IV has two limbs: its retention limb is the registered claim of proposition P15, word for word, and its comparison limb has no registered proposition yet; Law V carries propositions P12 and P21. Section 16 gives where each law stands. The note below was written before version 1.103 registered the two laws and is kept as printed; only its section pointer and its link are updated.

This paper and Laws IV and V, 3 October 2026. The theory has been stated with five laws since 24 September 2026. Law IV, the ARC Persistence Law, is carried by proposition P15, and Law V, the ARC Embedding Law, by propositions P12 and P21. The three propositions are registered. Neither law is registered as a law, neither has been tested, and their mathematics is set out in Laws IV and V: the mathematics, version 1.0, of 3 October 2026. Most of this paper concerns the forms and exponents of scaling laws in physical, biological and cosmological systems; none of that work measures the persistence of an installed gain, the order in which a system is built, or where correction is placed. Its summaries of the programme's AI experiments, in Section 12, relate capability and alignment scores to reasoning depth, and none of them measures those three things either. Where the paper turns to embedding, it states Prediction 5, that ethical reasoning loops embedded in an AI inference pipeline will improve alignment-relevant output quality. It reports the March 2026 pilot of the Eden Protocol, in which the Stakeholder Care Loop produced a statistically significant improvement in stakeholder care on all three working models, under cross-model scoring that was not blind. Those gains were measured with the loops in place. The pilot does not show what remains once the loops are removed or after later training, which is the question of P15; it does not vary the order in which a system is built, which is the question of P12; and it sets no correction that sees only the finished output against the loops, which is the comparison P21 makes as depth rises. No result in this paper is a deciding test of P15, P12 or P21. Naming the two laws changes nothing printed here: the results, their dates and their limits stand as they were.

D.3 Companions

This paper extends Paper I (*The ARC Equation, the Law of Conversion*, 17 January 2026) and is a companion to the Foundational Paper and to Paper VII (*Cauchy Unification: the Classification Behind the Law's Form*), its key companion.

Key companion: Paper VII · The ARC Theory · Law I, the ARC Principle · Cauchy Unification: the Classification Behind the Law's Form (50-domain empirical test of the Cauchy classification: 19/25 under the classification rule of March 2026; $p = 7 \times 10^{-5}$ by permutation, 1.56×10^{-5} binomial; no result from the analysis of 11 August 2026 is printed)

Other companions: Paper II, Paper III, Paper IV.a, Paper IV.b, Paper IV.c, Paper IV.d, Paper V, Paper VI, Paper VIII, Paper IX, Eden Engineering, Eden Vision, Executive Summary, Master Table of Contents.

D.4 Reading this paper

The full differential of this paper against every prior document of the programme is at Eden Vision, section II.A.8; the reading note of version 3.12 is kept in Appendix A.2.

In plain words. Earlier versions summed the paper up in these lines, kept here as printed: "One formula. From cells to cosmos."; "The discovery is in the connection, not the components."; and "One framework. From the heartbeat of a mouse to the expansion of the universe. And beyond both - to the scaling of thought." Read them with Section 16: the cosmos enters as an identity, which carries no evidence, and the scaling of thought is a different pair of variables, which has to be measured.

E Novelty statements and the closest earlier work

The dated correspondence note of 19 September 2026 (Appendix A.2) records that the paper's universal novelty statements were made with no literature search and that no novelty verdict is recorded either way. This appendix keeps each statement as printed, beside the closest earlier work the paper itself names, with its status in this version. A dated search remains to be made.

Table 19. Novelty statements.

Statement, as printed	Where	Closest earlier work the paper names	Status in this version
"To our knowledge, this explanation for why the two derivations converge has not been previously published. The observation that Cauchy's theorem, combined with space-filling and conservation, provides the meta-constraint underlying both programmes appears to be novel."	Section 12	West, Brown and Enquist (1997, 1999); Demetrius (2003, 2006); Demetrius and Tuszynski (2010); the review of Agutter and Tuszynski (2011); Banavar, Maritan and Rinaldo (1999)	kept, with a pointer here; no dated literature search recorded
The box label "Novel Result"	Section 12	the box itself gives the formula and its three values to the earlier work	withdrawn in this version: novelty is claimed only against a dated literature search, and none is recorded
"This has not been stated in this form." (on the space of classical laws as a three-parameter family)	Section 12 of version 3.12, seventh entry	Cauchy (1821)	withdrawn with the claim it qualified (the cardinality and scope note of 19 September 2026, Appendix A.2)
"the first time this transition has been identified as a change in composition operator"	Section 12 of version 3.12, ninth entry	none named	withdrawn with the claim it qualified (Appendix A.2)
"It has not been stated in this form." (the consequence for physics)	Section 15 of version 3.12	Cauchy (1821)	withdrawn with the claim it qualified (Appendix A.2)

The analogy, moved here from Section 2.2. Newton discovered that gravity exists and measured its effects. Einstein discovered *why*: geometry. Mass curves spacetime, and objects follow the curves. Newton was not wrong. He was incomplete.

Kleiber found that the basal metabolism of thirteen groups of animals, from ring doves to steers, is best calculated per 0.74 power of body weight, rather than per unit of body surface as the earlier surface law held. West, Brown, and Enquist derived the mechanism for three dimensions and gave the exponent for D dimensions. This paper's principle is the one Section 2.2 states. The likeness to Einstein's work is limited: Einstein's mechanism was compelled to fit a dated observational anomaly (Mercury's perihelion advance), whereas the four-cell grid here reorganises earlier derivations that were already correct in their own domains, and the empirical work that would settle it at law level (Section 11 and Section 16) is not yet run.